Automated Risk of Bias Assessment in Systematic Reviews Using Large Language Models: A Multi-Model Single-Agent Benchmark on 227 Randomised Controlled Trials

Luiz Felipe Guain Teixeira[1,2,3] Cassiano Pereira de Barros[4]
Renato Vicente[1,2,3]

[1]TELUS Digital Research Hub

[2]Center for Artificial Intelligence and Machine Learning, University of São Paulo

[3]Institute of Mathematics, Statistics and Computer Science, University of São Paulo

[4]Paulista School of Medicine, Federal University of São Paulo

{luiz.felipe.teixeira, rvicente}@usp.br

Abstract

Background: Risk of bias (RoB) assessment is a mandatory but labour-intensive step in systematic review production. Large language models (LLMs) offer the potential to automate this task in a zero-shot setting, yet systematic benchmarks spanning multiple frontier models and inference strategies are lacking.

Methods: Eight state-of-the-art LLMs (Gemini 2.5 Pro, Gemini 2.5 Flash, GPT-5.4, Claude Haiku 4.5, DeepSeek v4-Flash, Grok 4.3, Qwen 3.7-Plus, and GLM-5) were evaluated on zero-shot, single-agent RoB 1.0 assessment of 227 randomised controlled trials drawn from 127 Cochrane systematic reviews published in 2024. Models were accessed via the OpenRouter API in June 2026 using a fixed prompt and provider-default generation parameters (exact OpenRouter identifier strings in Table 2). Ground-truth annotations were taken from the published Cochrane consensus judgements across seven methodological domains (1,589 cells total). Performance was assessed using accuracy, Macro-F1, and Gwet's AC1, with study-clustered bootstrap 95% confidence intervals. Two models (Gemini 2.5 Flash and DeepSeek v4-Flash) were additionally evaluated under a single-call inference mode for comparison with the per-domain strategy.

Results: Overall accuracy ranged from 57.6% to 62.5% across models, a spread smaller than the width of individual 95% confidence intervals; Macro-F1 confidence intervals overlapped across all eight models. AC1 provided the clearest separation: Claude Haiku 4.5 (AC1 = 0.485; 95% CI $[0.443, 0.526]$) had non-overlapping CIs relative to Gemini 2.5 Pro ($[0.344, 0.420]$) and GPT-5.4 ($[0.349, 0.426]$), though this comparison is unadjusted and exploratory. Unclear Risk was the most consistently under-detected class (recall < 0.20 in three of seven domains across most models). Consensus failures (cells where no model was correct) accounted for only 14.2% of the benchmark, confirming substantial non-redundancy among models; the remaining 85.8% of cells had at least one correct model. The comparison of inference modes yielded small, largely non-significant differences: ten of twelve Δ metrics had confidence intervals spanning zero, and D7 (Other Bias)—the domain hypothesised to benefit most from joint assessment—showed negligible change under either mode.

Conclusions: Frontier LLMs can perform zero-shot RoB 1.0 assessment at accuracy levels broadly consistent with estimates of human inter-rater variability for this task [1], though direct comparison on this specific corpus was not performed; no single model dominates across all metrics, and no consistent advantage was found for single-call over per-domain inference. The primary practical distinction between inference modes is computational: single-call reduces inference calls by 7× without a reliable accuracy trade-off. All analyses are exploratory; results should be interpreted as directional benchmarks rather than definitive rankings.

Keywords: risk of bias; systematic review; large language models; natural language processing; Cochrane; evidence synthesis; automation

1 Introduction

Systematic reviews and meta-analyses occupy the apex of the evidence hierarchy in evidence-based medicine [2]. Their validity depends critically on the quality of the primary studies included: a meta-analysis that aggregates trials with uncontrolled confounding or selective outcome reporting inherits—and can amplify—those methodological flaws. The Cochrane Risk of Bias 1.0 tool (RoB 1.0) [3] was developed precisely to render this quality appraisal explicit and reproducible. By requiring reviewers to judge each included trial across seven pre-specified methodological domains, RoB 1.0 transforms a subjective quality impression into a structured, auditable record.

Despite this formalisation, RoB assessment remains a major bottleneck in systematic review production. Each trial must be read by two independent reviewers, who compare judgements and resolve disagreements through discussion or adjudication. For a review covering dozens of trials, this process routinely demands tens to hundreds of person-hours. Inter-rater reliability, even among trained reviewers, is moderate at best [4, 5]—a consequence of genuine ambiguity in reporting, domain-specific expertise requirements, and the epistemic character of the *Unclear Risk* category, which denotes insufficient information rather than a graded risk level. These limitations have motivated sustained interest in automating, or at least augmenting, the RoB assessment workflow.

Early automation efforts relied on rule-based systems and supervised machine learning trained on sentence-level text fragments [6]. Such approaches demonstrated that natural language processing could identify methodological descriptions in trial reports with above-chance accuracy, but they were constrained by limited contextual understanding, sensitivity to reporting style, and the need for substantial labelled training data for each domain.

Large language models (LLMs) trained on diverse corpora with instruction-following capabilities represent a qualitatively different paradigm [7, 8]. Without task-specific fine-tuning, general-purpose LLMs can be prompted to read a full trial report, locate domain-relevant passages, and produce structured judgements with free-text justifications—closely mimicking the expert reviewer workflow. A prompt template can address all seven RoB 1.0 domains either sequentially—one domain per inference call—or simultaneously in a single call; whether this choice affects assessment quality is an open empirical question. As frontier LLMs have grown more capable and their inference costs have fallen, evaluating their effectiveness for RoB assessment has become both feasible and practically important.

The present study provides a systematic empirical comparison of eight state-of-the-art LLMs on zero-shot, single-agent RoB 1.0 assessment applied to 227 randomised controlled trials drawn from 127 Cochrane systematic reviews published in 2024. All models were evaluated on a common benchmark under identical conditions: the same prompt, the same fully text-based study representations produced by a standardised PDF-to-Markdown pipeline, and the same ground-truth human annotations. We report accuracy, Macro-F1, and Gwet's AC1—metrics selected to characterise both aggregate performance and class-level sensitivity without imposing an ordinal interpretation on the three nominal RoB categories. Stratified analyses by intervention type and clinical domain identify sources of systematic variation across the benchmark. Additionally, we compare two inference strategies—per-domain and single-call—using Gemini 2.5 Flash and DeepSeek v4-Flash, the two models evaluated under both conditions, to assess whether querying all seven domains simultaneously affects performance relative to sequential per-domain calls.

2 Methods

2.1 Dataset

The ground-truth dataset was constructed from Cochrane systematic reviews published in 2024. Of the 149 reviews published that year, 127 were accessible through our institutional subscription; from each accessible review, up to three included randomised controlled trials (RCTs) were selected on the basis of full-text PDF availability: all included RCTs for which a PDF was retrievable were included, up to a maximum of three per review, yielding a final corpus of **227 RCTs** drawn from 127 distinct Cochrane reviews.

Risk of bias annotations were obtained directly from the published Cochrane review reports and reflect the consensus judgements produced by the original review teams following standard Cochrane procedures (independent assessment by at least two trained reviewers, with disagreements resolved by discussion or arbitration). These annotations were not re-produced or independently adjudicated for the present study; they represent the published Cochrane consensus, which constitutes the reference standard used by the field. Although the revised Risk of Bias 2.0 tool (RoB 2; Sterne et al. 9) is now recommended for new systematic reviews, it reorganises the domain structure and changes the rating scale, rendering it incompatible with the existing RoB 1.0 corpus. RoB 1.0 remains the basis for the large majority of existing systematic review annotations — including all 127 reviews sampled here and most of the large labelled datasets used for training and evaluation of automated systems — so this study's focus on RoB 1.0 is both corpus-driven and practically motivated. Each study was annotated using the Cochrane Risk of Bias 1.0 (RoB 1.0) tool [3], which covers seven methodological domains: D1 (random sequence generation), D2 (allocation concealment), D3 (blinding of participants and personnel), D4 (blinding of outcome assessment), D5 (incomplete outcome data), D6 (selective outcome reporting), and D7 (other sources of bias). Each domain received one of three judgements—*Low Risk*, *High Risk*, or *Unclear Risk*—yielding 1,589 annotated cells (227 × 7).

Class prevalence varied substantially across domains (Table 1). Low Risk was the largest class in six of seven domains, and the majority class (>50%) in four of them (D1, D5, D6, D7); only D3 (blinding of participants and personnel) had High Risk as its largest class (45%), reflecting the predominance of unblinded pharmacological trials in the corpus. At the skewed extreme, D1 had a 69% Low Risk majority and only 5 High Risk cases (2%); D5 and D7 were similarly imbalanced at 63% and 61%

Low Risk, respectively. D4 (blinding of outcome assessors) was the most balanced domain overall (44% Low Risk, 28% High Risk, 29% Unclear Risk), closely followed by D2, in which Low Risk and Unclear Risk were nearly equally represented (50% vs. 46%).

Table 1: Ground-truth class distribution per RoB domain (n = 227 studies per domain). LR = Low Risk; HR = High Risk; UR = Unclear Risk.

Domain	LR	HR	UR	Majority class	Majority (%)
D1	157	5	65	Low Risk	69.2
D2	113	10	104	Low Risk	49.8
D3	74	103	50	High Risk	45.4
D4	99	63	65	Low Risk	43.6
D5	143	39	45	Low Risk	63.0
D6	125	25	77	Low Risk	55.1
D7	139	31	57	Low Risk	61.2
Total	**850** (53.5%)	**276** (17.4%)	**463** (29.1%)	Low Risk	53.5

Intervention type. Each study was assigned a single intervention category: drug/medication (n = 104), non-pharmacological intervention (n = 75), procedure (n = 30), or medical device (n = 18).

Clinical domain. Studies were tagged with one or more of 24 Cochrane Review Group categories (multi-label; 120 of 227 studies belonged to more than one group). The most frequent categories were Child Health (n = 72), Heart & Circulation (n = 45), and Kidney Disease (n = 31); the least frequent were Skin Disorders (n = 3), Effective Practice & Health Systems (n = 4), Rheumatology (n = 5), and Wounds (n = 5). For stratified analyses, each multi-domain study was included in every applicable group.

2.2 Document Preparation Pipeline

Full-text study reports (PDF) were processed through a two-stage pipeline designed to produce fully text-based study representations consumable by any large language model without multimodal (image) input.

Stage 1 — PDF-to-Markdown conversion. Each PDF was converted to

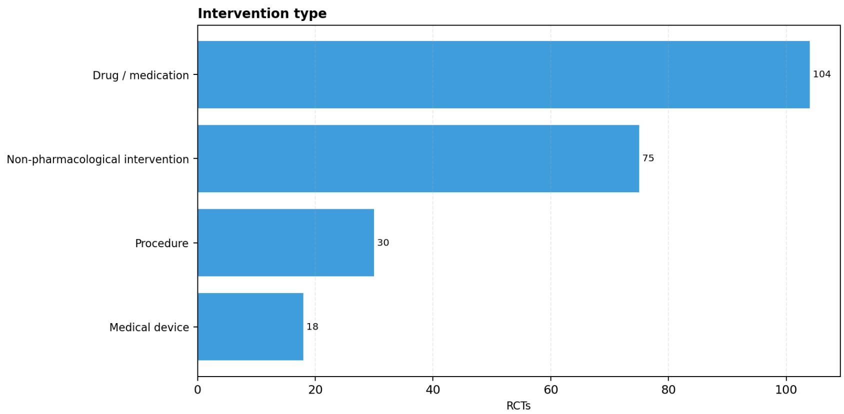

Figure 1: Distribution of the 227 RCTs by intervention type.

structured Markdown using MinerU [10], which preserves headings, tables, and inline text whilst replacing embedded figures with placeholder tokens of the form `[Figure` k`]`.

Stage 2 — Figure description. Each figure placeholder was resolved by a vision-language model (VLM) that received the corresponding page image and generated a textual description, which was then inserted in place of the placeholder. After Stage 2, the document was a self-contained Markdown file in which all information—including the content of charts, CONSORT flow diagrams, Kaplan–Meier curves, and forest plots—was represented as plain text. This design eliminates the need to pass images alongside the prompt during RoB assessment, enabling evaluation of text-only models and ensuring a consistent input format across all eight models tested.

2.3 Risk of Bias Assessment

All models were evaluated in a zero-shot, single-agent setting: each processed Markdown document was submitted to the model with a fixed assessment prompt (reproduced in full in Supporting Information S1), which specified the domain criteria, the three admissible judgements (Low Risk, High Risk, Unclear Risk), and the required JSON output format. No few-shot examples were provided, and the prompt was used without modification across all models.

Inference modes. We evaluated two inference strategies that differ in how the seven RoB domains are requested from the model. In the *per-domain* mode, seven

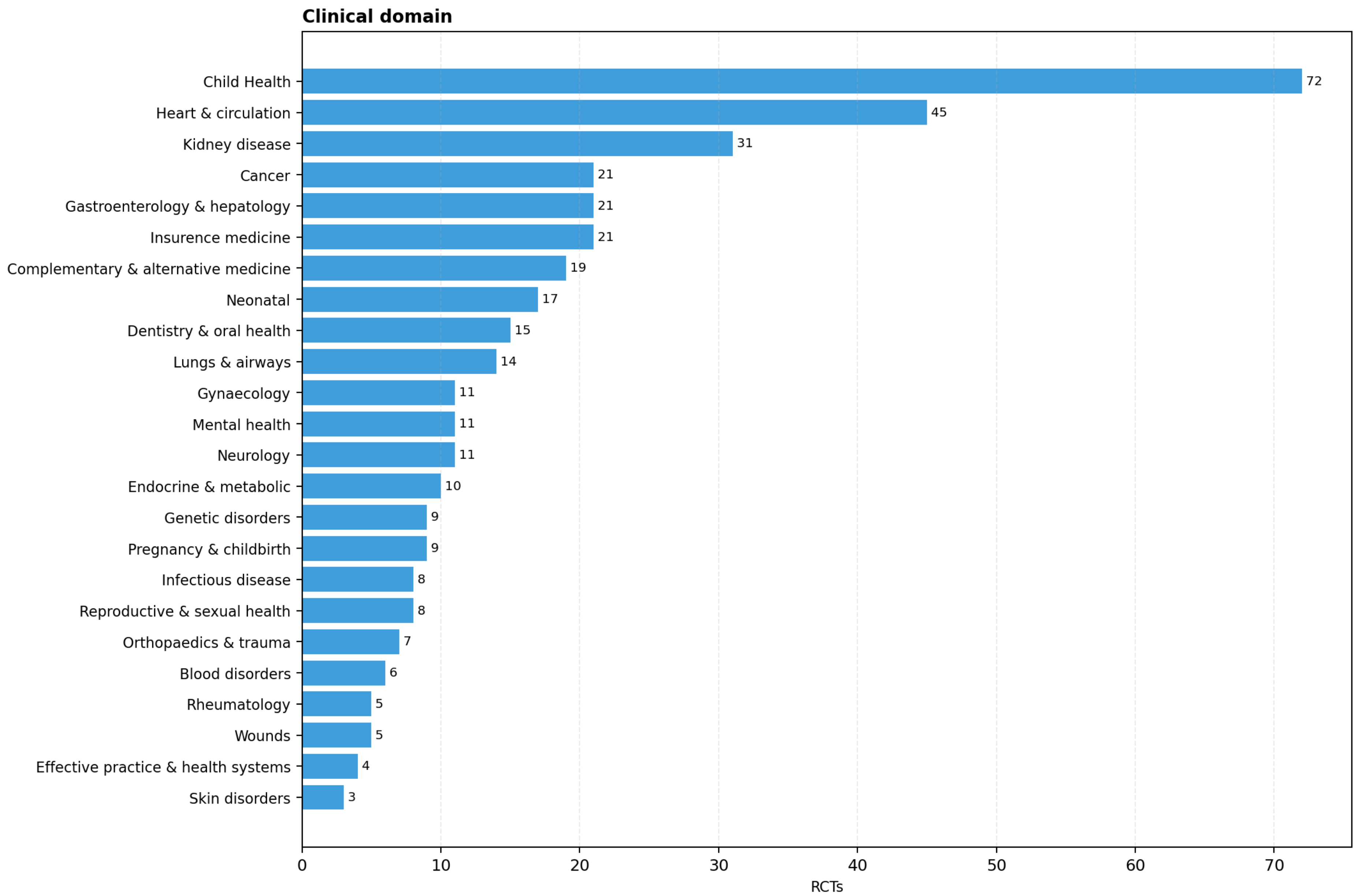

Figure 2: Distribution of the 227 RCTs by clinical domain (Cochrane Review Group categories, multi-label; total exceeds 227).

sequential inference calls are made per study: each call supplies the full Markdown text together with a single domain identifier (e.g. `Domain: D1`), and the model returns a JSON object containing a judgement and justification for that domain only. In the *single-call* mode, a single inference call per study is made: the model receives the full Markdown text and is instructed to return a JSON array containing judgements and justifications for all seven domains simultaneously. All eight models were evaluated under the per-domain mode. Gemini 2.5 Flash and DeepSeek v4-Flash were additionally evaluated under the single-call mode on the same 227 studies, enabling a controlled comparison of the two inference strategies whilst holding model, prompt content, and corpus constant.

2.4 Language Models

Eight frontier LLMs were evaluated (Table 2), accessed via the OpenRouter API in June 2026 using the identifier strings listed therein. No model-specific fine-tuning, prompt engineering beyond the fixed assessment prompt (Section 2.3), or retrieval augmentation was applied.

Evaluation protocol. All calls used provider-default generation parameters and each study–condition pair received a single inference draw with no output aggregation, constituting a hyperparameter-free single-draw evaluation protocol. This design reflects two constraints. First, decoding parameters such as temperature interact with model architecture and training objective in model-family-specific ways; varying them across models would confound generation configuration with prompt design, rendering cross-model comparisons uninterpretable. Second, post-hoc reliability strategies—including self-consistency decoding [11], output ensembling, and majority voting over repeated draws—are optimisations orthogonal to native zero-shot capability and constitute a separate experimental question outside the scope of this benchmark.

Reproducibility and versioning. OpenRouter routes each identifier string to the provider-current checkpoint at inference time; underlying weights may be updated by the provider without a corresponding change to the public identifier. All results are therefore conditioned on the checkpoint states active during the June 2026 inference window and should not be assumed to generalise to subsequent revisions of the same identifier strings.

Table 2: Language models evaluated. All models accessed via the OpenRouter API in June 2026 using the model identifier strings shown; zero-shot single-agent mode throughout. Context length: maximum tokens accepted per call as reported by OpenRouter at access time. Release date: date the model became available on OpenRouter. Exact checkpoint versions are opaque at the provider level; the identifiers below are the strings passed to the API and represent the model state at the time of inference.

Model	OpenRouter identifier	Context	Release	Studies
Gemini 2.5 Pro	`google/gemini-2.5-pro`	1M	Jun 2025	227
Gemini 2.5 Flash	`google/gemini-2.5-flash`	1M	Jun 2025	227
GPT-5.4	`openai/gpt-5.4`	1M	Mar 2026	227
Claude Haiku 4.5	`anthropic/claude-haiku-4.5`	200K	Oct 2025	227
DeepSeek v4-Flash	`deepseek/deepseek-v4-flash`	1M	Apr 2026	227
Grok 4.3	`x-ai/grok-4.3`	1M	Apr 2026	227
Qwen 3.7-Plus	`qwen/qwen3.7-plus`	1M	May 2026	227
GLM-5	`z-ai/glm-5`	203K	Feb 2026	227

2.5 Evaluation Metrics

Overall accuracy. The proportion of cells in which the model's judgement matched the ground-truth label, computed across all 1,589 study–domain pairs.

Per-class F1. For each class $k \in \{\text{Low Risk, High Risk, Unclear Risk}\}$, the F1 score is the harmonic mean of precision and recall:

$$\mathrm{F1}_k = \frac{2 \cdot \mathrm{Precision}_k \cdot \mathrm{Recall}_k}{\mathrm{Precision}_k + \mathrm{Recall}_k}, \tag{1}$$

computed across all study–domain cells. Per-class F1 reveals class-specific failure modes independently of overall performance.

Macro-F1. Per-class F1 is computed over the full pool of 1,589 cells; the three class-level F1 scores are then averaged with equal weight: $\text{Macro-F1} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{F1}_k$ with $K = 3$. Macro-F1 assigns equal weight to each category regardless of prevalence, thereby penalising models that achieve high accuracy by concentrating predictions on majority classes.

Gwet's AC1. A chance-corrected agreement coefficient robust to the prevalence paradox [12, 13]: when marginal distributions are highly skewed, classical chance-corrected coefficients approach zero even when observed agreement is substantial. AC1 is defined as

$$\mathrm{AC1} = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = \frac{1}{K-1}\sum_{k=1}^{K} \pi_k(1-\pi_k), \tag{2}$$

where p_o is the observed proportion of agreement, $K = 3$ is the number of categories, and $\pi_k = \left(\hat{p}_{k,\mathrm{model}} + \hat{p}_{k,\mathrm{gt}}\right)/2$ is the mean marginal proportion for category k, with $\hat{p}_{k,\mathrm{model}}$ and $\hat{p}_{k,\mathrm{gt}}$ the marginal proportions of class k in the model predictions and ground truth, respectively. All metrics—including AC1—are computed globally by pooling all 1,589 study–domain cells; domain-level values are not aggregated. Given the pronounced class imbalance documented in Table 1—six of seven domains have Low Risk as the majority class—AC1 is more appropriate than Cohen's κ as the primary agreement metric for this task, since κ approaches zero under heavy prevalence skew even when observed agreement is substantial (the prevalence paradox; Feinstein and Cicchetti 13). Importantly, the ground-truth annotations used here reflect published Cochrane consensus judgements rather than a second independent annotator; AC1 therefore quantifies model-to-published-standard agreement, not inter-rater reliability in the classical sense. Human inter-rater reliability for RoB 1.0 is moderate at best: even among formally trained reviewers applying the same protocol, Hartling et al. [1] reported $\kappa = 0.79$ for D1 but only $\kappa = 0.24$–0.37 for the remaining domains, with agreement between independent consensus pairs falling further to $\kappa = 0.05$–0.09 in several domains. Agreement also varies substantially between distinct Cochrane review teams evaluating the same trials [4]. These findings contextualise model performance against a noisy reference standard rather than a true human ceiling, and motivate domain-level rather than aggregate interpretation of results.

Consensus failures and model complementarity. For each study–domain cell, we recorded whether at least one of the eight models matched the ground truth. Cells where *no* model was correct are termed *consensus failures* and represent cases in which all evaluated models failed simultaneously, regardless of which model was consulted. The *any-correct rate* (proportion of cells where at least one model succeeded) characterises the degree of complementarity among models: a high any-correct rate alongside low individual-model accuracy indicates that different models fail on different cells, yielding non-redundant coverage across the ensemble.

Stratified analyses. All primary metrics were recomputed within subgroups defined by intervention type (four categories) and clinical domain (24 Cochrane Review Groups, multi-label). For clinical domain, studies were included in every group to which they belonged, so group sample sizes sum to more than 227.

Statistical inference. All analyses are descriptive and exploratory. Confidence intervals are reported to characterise estimation uncertainty and facilitate comparison across models and conditions; they are not interpreted as the outcome of pre-specified confirmatory hypothesis tests. No familywise error rate correction was applied: the number of comparisons across models, domains, classes, and inference modes is large and the outcomes are correlated, making conventional multiplicity corrections either too conservative (Bonferroni) or difficult to calibrate without a pre-specified primary endpoint. Where 95% bootstrap CIs exclude zero, this is noted descriptively ("CI excludes zero"); the 95% level serves as a conventional descriptive reference threshold and carries no confirmatory implication in this exploratory context—it marks an unadjusted signal that may warrant further investigation, not a principled significance claim.

3 Results

3.1 Overall Performance

Figure 3 summarises accuracy, Macro-F1, per-class F1, and Gwet's AC1 for all eight models across the 1,589 annotated cells, together with study-clustered bootstrap 95% confidence intervals (5,000 iterations, resampling the 227 studies—the unit of clustering—with replacement to preserve within-study correlation across the seven domains). Overall accuracy ranged from 57.6% (Gemini 2.5 Pro) to 62.5% (Claude Haiku 4.5), a spread of less than five percentage points; the 95% CI for each model spans approximately 5–6 pp, so all pairwise accuracy differences fall within sampling uncertainty and should be treated as directional. Macro-F1 confidence intervals similarly overlap across all eight models, precluding reliable rank ordering based on this metric alone. By contrast, Gemini 2.5 Flash and DeepSeek v4-Flash tied for the highest Macro-F1 (0.558) while Claude Haiku 4.5—the accuracy leader—ranked sixth (0.541). The per-class breakdown reveals that the Macro-F1 gap is driven primarily by High Risk and Unclear Risk detection: Claude Haiku 4.5 achieved the highest Low Risk F1 (0.743) but the lowest High Risk F1 (0.469) and Unclear Risk F1 (0.413) among all models. Among the three metrics, Gwet's AC1 shows the clearest separation at the extremes: the 95% CI for Claude Haiku 4.5 (AC1 = 0.485; CI $[0.443, 0.526]$) does not overlap with those of Gemini 2.5 Pro ($[0.344, 0.420]$) or GPT-5.4 ($[0.349, 0.426]$), indicating that Haiku's higher chance-corrected agreement is not attributable to sampling variation. Consensus failures—cells where no model in the ensemble was correct—accounted for 226 of 1,589 cells (14.2%); the complementary any-correct rate of 85.8% confirms that the eight models make substantially non-redundant errors (Section 3.8).

Metric divergence. The discrepancy between accuracy and Macro-F1 rankings reflects differential handling of the class imbalance documented in Table 1. Figure 3 makes the trade-off explicit: models with higher Low Risk F1 tend to achieve higher accuracy but lower High Risk and Unclear Risk F1, since Low Risk is the majority

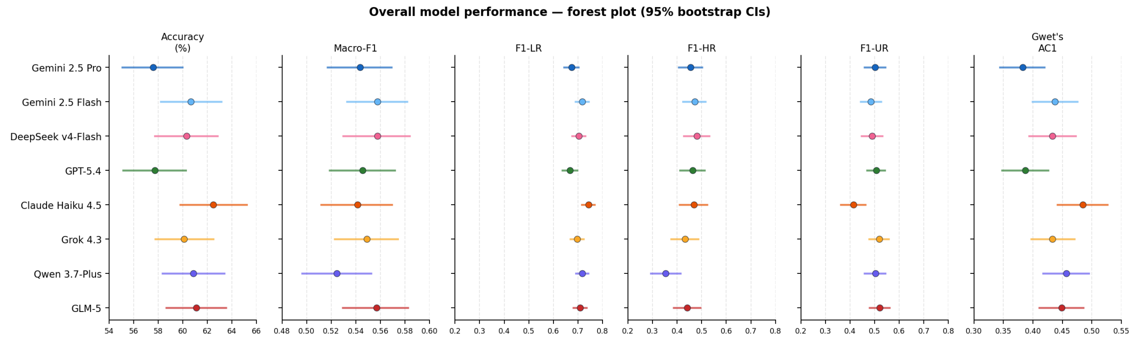

Figure 3: Overall performance on RoB 1.0 assessment for the eight models ($n = 227$ studies; 1,589 cells). Each row is one model; columns show accuracy (%), Macro-F1, F1 for each risk class (F1-LR, F1-HR, F1-UR), and Gwet's AC1. Filled circles are point estimates; horizontal bars are 95% study-clustered bootstrap confidence intervals (5,000 iterations, resampling 227 studies with replacement). Dashed vertical lines mark the cross-model mean for each metric. Colours identify models consistently throughout the paper. Full numerical values are reported in Table SS1.

class in six of seven domains. Macro-F1 penalises this behaviour by weighting each class equally, shifting the ranking toward models with more balanced per-class detection. Inspection of Figure 3 also reveals that the rank orderings for accuracy, AC1, and F1-LR are nearly identical across the eight models. This co-linearity suggests that all three metrics are largely co-determined by Low Risk performance: a model that reliably assigns Low Risk to majority-class cells captures most of the variance in both accuracy and chance-corrected agreement, since Low Risk prevalence (53.5%) dominates both the observed agreement count and the marginal distributions that enter the AC1 correction. Macro-F1 disrupts this co-linearity by treating each class as equally important regardless of prevalence. However, equal class weighting does not reflect the empirical frequency of judgements—High Risk accounts for only 17.4% of cells (Table 1)—so macro-F1 amplifies the contribution of the rarest class beyond its share of the data. A prevalence-weighted F1 would instead converge toward accuracy, whereas macro-F1 should be interpreted as a measure of balanced class coverage: how well a model performs across *all three* nominal categories, independently of how often each arises.

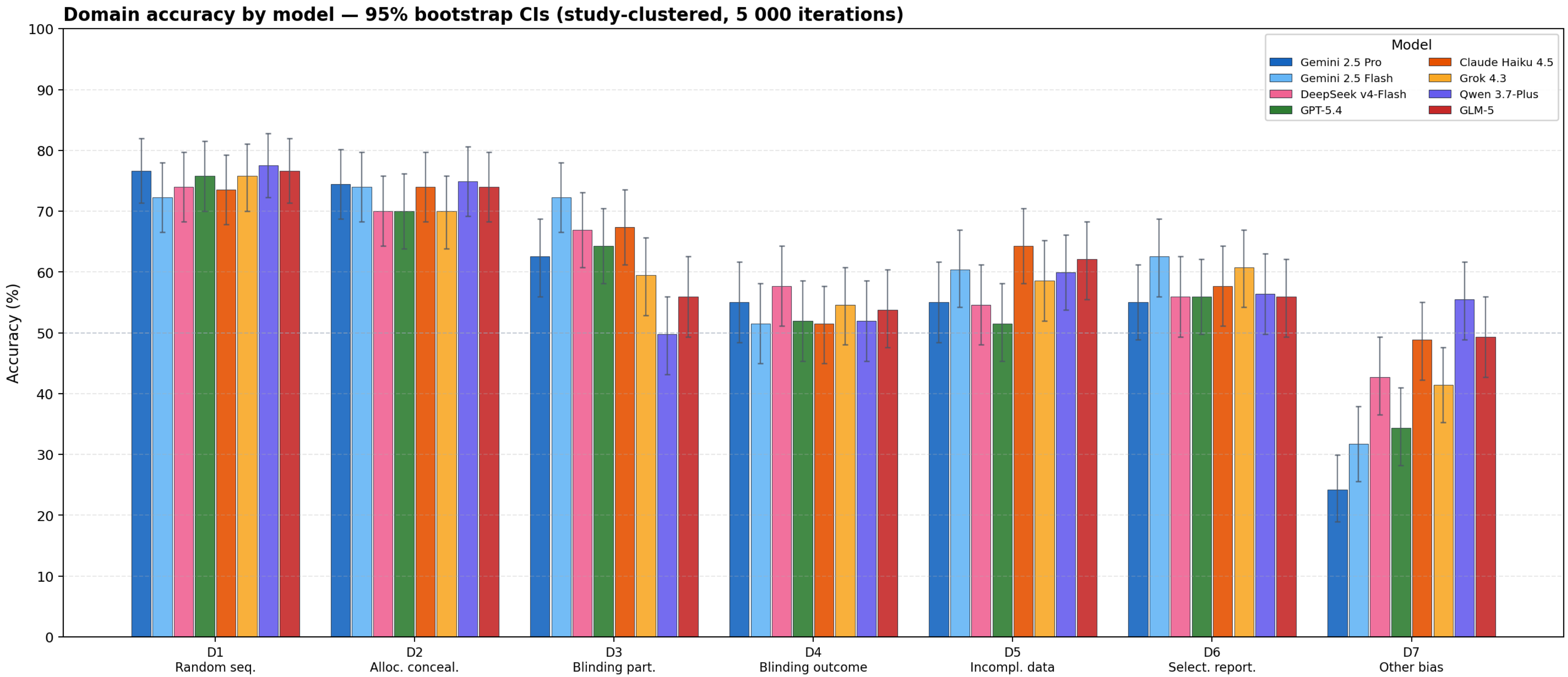

Figure 4: Per-domain accuracy by model (95% bootstrap CIs; study-clustered, 5,000 iterations, resampling 227 studies with replacement). Each group of eight bars corresponds to one RoB domain; colours identify models as shown in the legend. The dashed horizontal line marks 50% (majority-class baseline). All analyses are exploratory. See Figure 5 for the breakdown by Macro-F1, per-class F1, and Gwet's AC1.

3.2 Performance by RoB Domain

Domain-level accuracy (Figure 4; Table SS2) revealed two tiers of difficulty. **D1** (sequence generation) and **D2** (allocation concealment) were the most accurately assessed domains, with mean accuracy of 75.3% and 72.7%, respectively; all models exceeded 70% accuracy on D1. **D3** (blinding of participants) showed high variance across models (49.8%–72.2%), with Gemini 2.5 Flash and Claude Haiku 4.5 outperforming others by a margin of more than 5 percentage points.

D7 (other sources of bias) was the most challenging domain for all models, with accuracy ranging from 24.2% (Gemini 2.5 Pro) to 55.5% (Qwen 3.7-Plus)—a spread of 31 percentage points within a single domain. Notably, models at opposite ends of the D7 accuracy range adopted diametrically opposed prediction strategies: Gemini 2.5 Pro assigned High Risk to over 93% of D7 cells (recall 0.94 for High Risk; recall 0.18 for Low Risk), whereas Qwen 3.7-Plus assigned Low Risk to 88% of D7 cells (recall 0.88 for Low Risk; recall 0.06 for High Risk). Both strategies yielded similarly poor Macro-F1 in D7 (≤ 0.37 for both), demonstrating that high accuracy on a single class does not translate to overall competence.

Consensus failures were not uniformly distributed across domains: the consensus failure rate was lowest for D2 (8.8% of its 227 cells) and highest for D4 (22.0%), consistent with the greater class balance and annotation difficulty in D4 relative to D2.

The domain-level pattern identifies two structurally distinct types of difficulty. **D4** (blinding of outcome assessors) exhibits systematic, directionally consistent errors: across all eight models, the large majority of errors are lenient—models assign Low Risk when the ground truth is High Risk (see Figure 6)—a pattern present across all intervention types. In drug trials in particular, models appear to conflate the objectivity of the outcome measure with the adequacy of blinding of the outcome assessor, defaulting to Low Risk when no explicit blinding failure is stated. **D7** (other sources of bias), by contrast, is characterised by inter-model divergence: models adopt idiosyncratic strategies—some biased toward High Risk, others toward Low

Risk—so that errors are distributed roughly equally between lenient and cautious directions. The wide accuracy spread across models in D7 (31 pp) and its near-zero chance-corrected agreement thus reflect not a shared systematic error but genuine criterion ambiguity; the residual difficulty in D7 is unlikely to be resolved by prompt refinement alone.

3.3 Per-class Analysis

Figure 5 disaggregates Macro-F1 into its three constituent classes. Across all models, the ordering F1-LR $\gg$ F1-UR $\geq$ F1-HR holds as a near-universal pattern, closely mirroring the prevalence ordering of the three classes (53.5%, 29.1%, and 17.4%, respectively; Table 1). Models detect each class roughly in proportion to how often they encounter it in training and inference, producing a performance gradient that tracks class frequency. The sole exception is Claude Haiku 4.5, which inverts the UR–HR ordering (F1-HR $= 0.469 >$ F1-UR $= 0.413$), suggesting a different response calibration: Haiku is more willing to assign High Risk when text signals a methodological concern, at the cost of under-using Unclear Risk as an epistemic hedge—a pattern consistent with its highest F1-LR (0.743) and lowest F1-UR (0.413) in the entire benchmark.

Low Risk detection was strong and consistent across models in D1 and D2 (F1 > 0.75 for all models). Performance deteriorated in D7, where most models reported Low Risk F1 below 0.60; Gemini 2.5 Pro collapsed to F1 $= 0.29$ in D7 as a consequence of its High Risk-biased prediction strategy described above.

High Risk detection was reliable only in D3 (F1: 0.47–0.79), the single domain in which High Risk cases outnumbered Low Risk cases in the ground truth (103 vs. 74; Table 1). The wide bootstrap confidence intervals for F1-HR visible in D1 and D2 in Figure 5 are a direct consequence of the small number of High Risk cases in those domains (D1: $n = 5$; D2: $n = 10$ of 227 studies), such that a single misclassified study shifts the F1 estimate substantially under resampling; the point estimates themselves carry accordingly low precision and should not be compared across models in those domains. In D6 (selective outcome reporting), four models reported High Risk F1 = 0.00 (Gemini 2.5 Flash, GPT-5.4, Claude Haiku 4.5, and Qwen 3.7-Plus), corresponding to zero true-positive High Risk predictions across all 227 studies in that domain. High Risk is the least prevalent class in D6 (25 of 227 cases, 11%; Table 1), and these models predicted fewer than three High Risk cases in D6 each, systematically defaulting to Low Risk or Unclear Risk in the absence of explicit evidence of selective reporting.

Unclear Risk was the most consistently under-detected class. Mean recall across all models fell below 0.20 in D3, D5, and D7. The worst single result was Claude Haiku 4.5 in D5: recall = 0.04, meaning the model correctly identified only 2 of the 45 Unclear Risk cases in D5. The most accurately detected Unclear Risk domain was D2, where all models achieved F1 > 0.70, consistent with the near-equal prevalence of Low Risk and Unclear Risk judgements in that domain (113 vs. 104).

Unclear Risk as abstention. The pattern of Unclear Risk under-detection reflects an asymmetric decision behaviour. In D3, D4, D5, and D7—domains where the ground-truth Unclear Risk rate ranges from 20% to 29%—all models used Unclear Risk substantially less often than the Cochrane reviewer, committing to definitive Low Risk or High Risk judgements even in cases where the reviewer judged methodological information to be insufficient. In D2 and D6, the pattern reverses: models over-used Unclear Risk relative to the ground truth, reflecting excessive hedging in the face of ambiguous reporting. This bidirectional discrepancy indicates that models calibrate their uncertainty threshold differently across domains and that reducing false positives for Unclear Risk in the blinding domains (D3–D4) is a necessary condition for improving High Risk recall there.

3.4 Error Direction: Leniency and Caution

Among cells where models erred, errors were not directionally neutral. Figure 6 shows the percentage of errors that were lenient across domains and intervention types.

D4 (blinding of outcome assessors) exhibited the most pronounced and consistent leniency, with lenient errors exceeding 70% in drug/medication trials across almost all models. This is interpretable: in pharmacological trials, the use of double-blind placebo-controlled designs is common, and models may be defaulting to Low Risk when such language is absent, rather than correctly inferring High Risk from the lack of an explicit blinding protocol.

D3 (blinding of participants and personnel) showed a gradient by intervention type: drug trials showed lower leniency (blinding is frequently achieved), whereas device and procedural trials showed the highest leniency rates, consistent with the fact that physical intervention concealment is often impossible in those settings yet models continued to default to Low Risk.

D1 (sequence generation) and **D2** (allocation concealment) showed approximately balanced error directions across all models and intervention types, consistent with their higher accuracy and the availability of explicit methodological language in well-reported trials. **D7** (other sources of bias) showed no dominant direction, with models diverging between lenient and cautious errors—confirming the heterogeneous character of errors in this domain noted in Section 3.2.

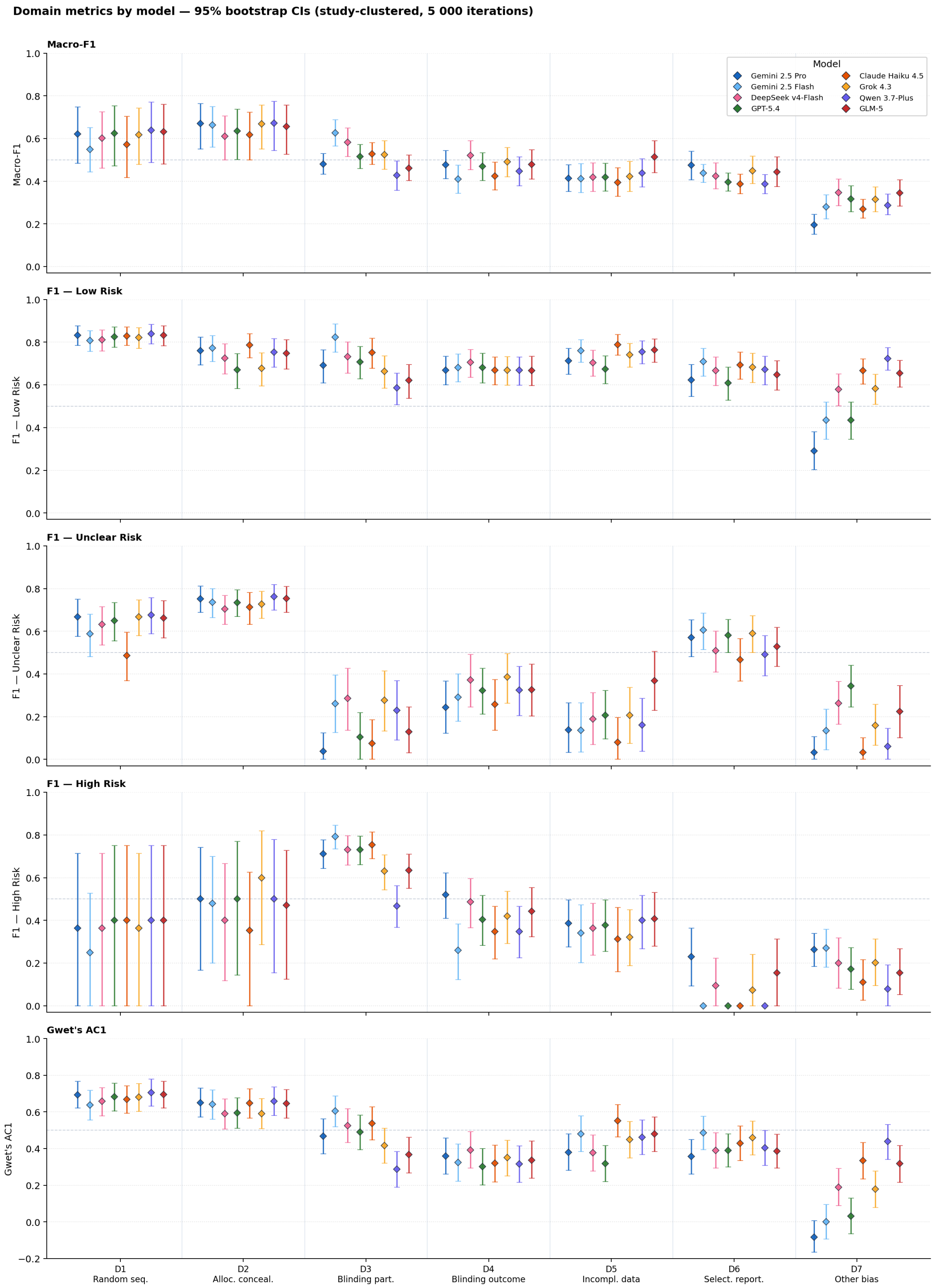

Figure 5: Per-domain performance by model across five metrics: Macro-F1, F1 for Low Risk, Unclear Risk, and High Risk, and Gwet's AC1 (95% bootstrap CIs; study-clustered, 5,000 iterations, resampling 227 studies with replacement). Each panel shares the same domain grouping on the x-axis; diamonds are point estimates; vertical bars are 95% CIs. The dashed horizontal line marks 0.5. Colours identify models consistently throughout the paper.

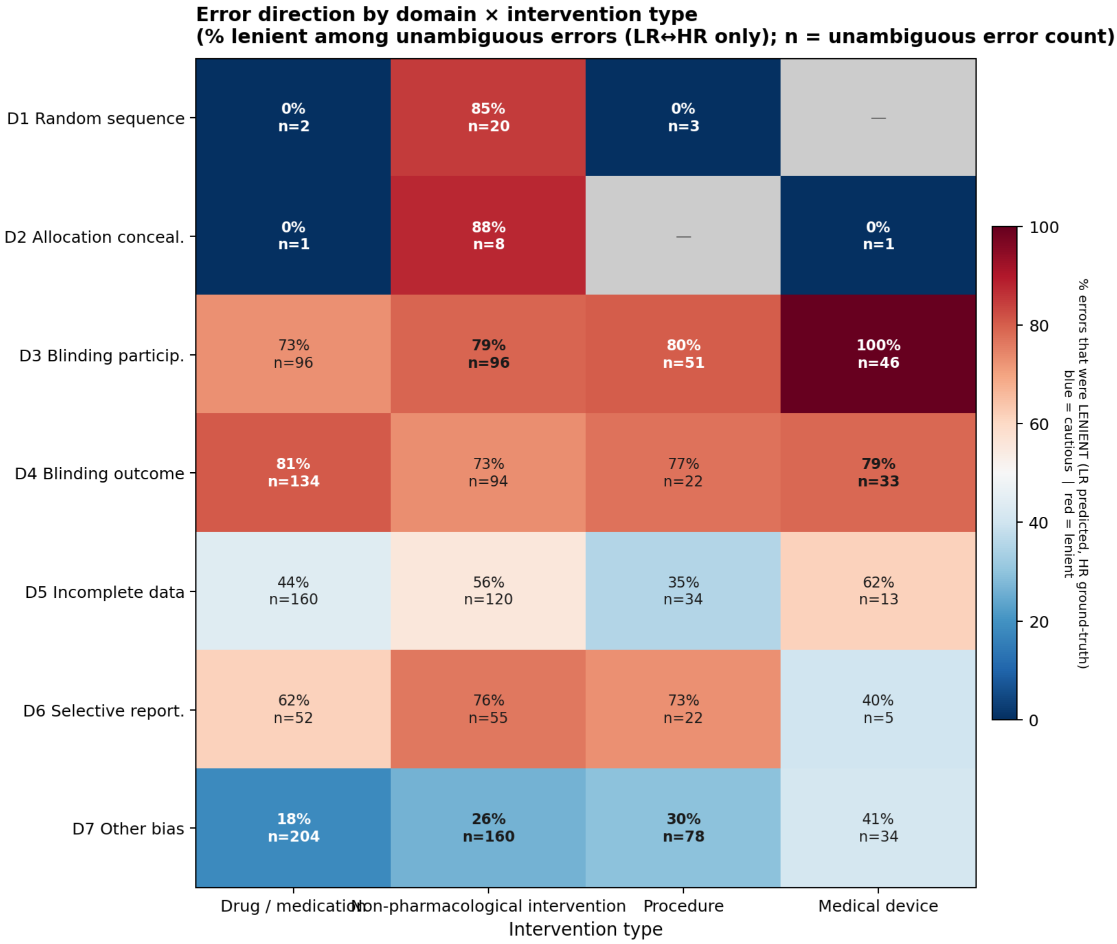

Figure 6: Error direction by RoB domain and intervention type. Each cell shows the percentage of errors that are lenient (model assigns a lower-risk class than the ground truth) under an ordinal scale Low Risk < Unclear Risk < High Risk; n = total errors in each cell. Values above 50% indicate a lenient bias; values below 50% indicate a cautious bias.

3.5 Stratification by Intervention Type

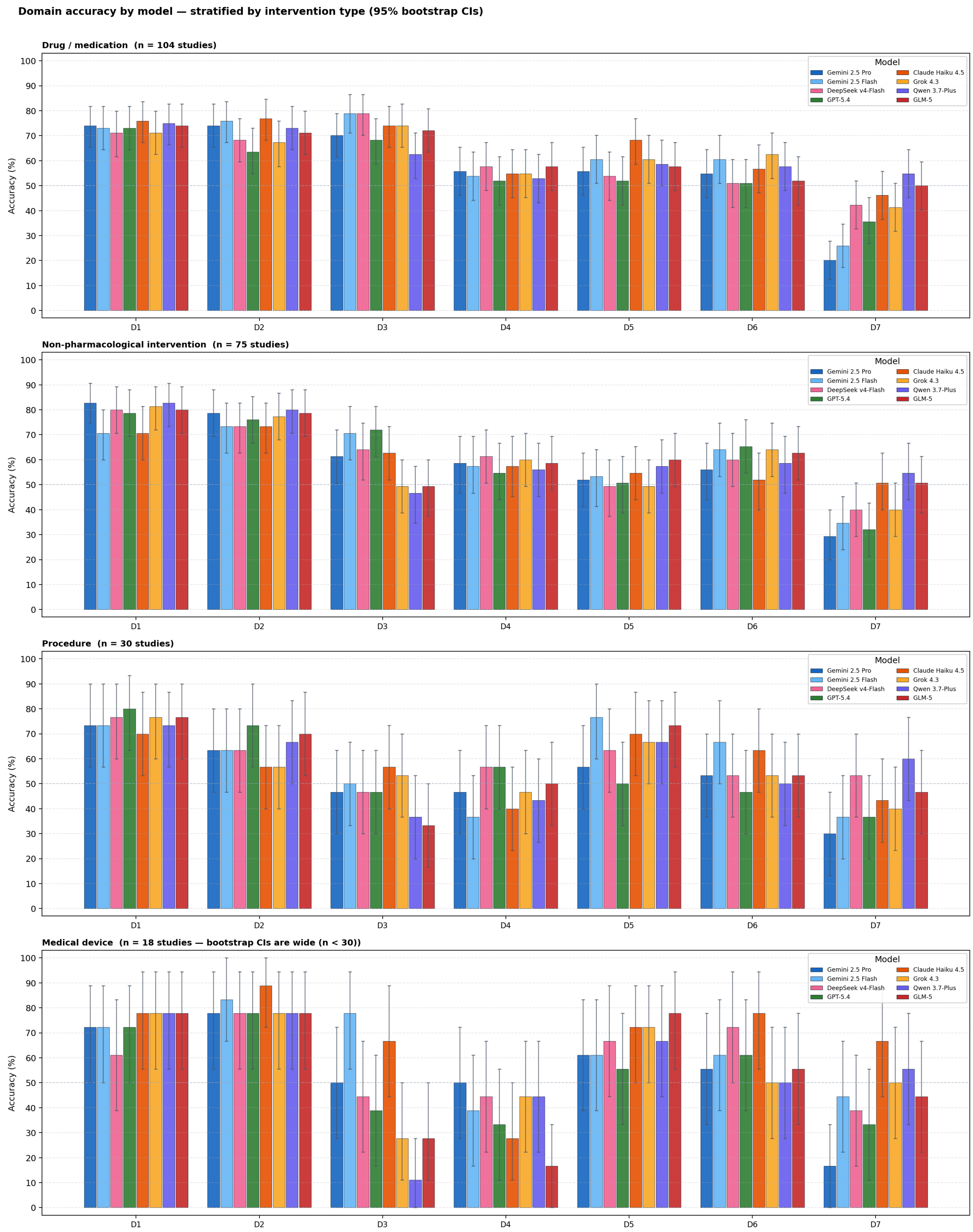

Figure 7: Per-domain accuracy by model, stratified by intervention type (95% bootstrap CIs; study-clustered, 5,000 iterations). Panels from top to bottom: drug/medication ($n = 104$), non-pharmacological intervention ($n = 75$), procedure ($n = 30$), and medical device ($n = 18$). Bootstrap CIs for the two smaller strata are wide and should be interpreted with caution; see Supporting Information S4 for numeric estimates.

Mean Macro-F1 across all models ranked intervention types as follows: non-pharmacological intervention (0.572, $n = 75$ studies), drug/medication (0.538, $n = 104$), medical device (0.531, $n = 18$), and procedure (0.510, $n = 30$). The 6.2-point gap between non-pharmacological interventions and procedures was consistent across most models; no single model reversed this ordering. Individual model rankings within each intervention type are shown in Figure 7 and provided in full in Table SS4.

The interaction between intervention type and blinding assessment reveals a specific failure mode. In **D3** (blinding of participants and personnel), accuracy for drug/medication trials substantially exceeded that for device and procedural trials; in the latter, the ground-truth High Risk prevalence is highest—because physical concealment of the intervention is frequently impossible—yet models assigned Low Risk to the majority of cells, producing leniency rates above 65% (Figure 6). In **D4** (blinding of outcome assessors), the leniency bias reached its maximum in drug trials (exceeding 80%), where models appear to assume that objective outcome measurement implies adequate assessor blinding. These findings converge on a specific prompt deficiency: the current zero-shot prompt does not distinguish between trial types when applying blinding criteria. Models apply a "blinding is feasible and likely" default regardless of the physical feasibility of concealment, an error that a single conditional rule in the prompt (e.g., flagging as High Risk by default when concealment is structurally impossible and no blinding mechanism is described) would substantially reduce.

3.6 Stratification by Clinical Domain

Among clinical domains with at least five studies, mean Macro-F1 across models ranged from 0.425 (Infectious Disease, $n = 8$) to 0.680 (Orthopaedics & Trauma, $n = 7$)—a 25.5-point spread (Figure 8). The four highest-performing domains were Orthopaedics & Trauma (0.680), Endocrine & Metabolic (0.638, $n = 10$), Rheumatology (0.637, $n = 5$), and Wounds (0.628, $n = 5$). The four lowest-performing domains were Infectious Disease (0.425), Reproductive & Sexual Health

(0.437, $n = 8$), Genetic Disorders (0.473, $n = 9$), and Mental Health (0.481, $n = 11$). Complete results for all 24 Cochrane Review Groups are provided in Table SS5.

These results are hypothesis-generating rather than confirmatory. Group sizes of 5–11 studies at the extremes preclude reliable rank-ordering across domains, and multi-label tagging makes groups non-exclusive (some reviews are cross-listed under multiple Cochrane Review Groups). The pattern is nonetheless qualitatively consistent with an interpretable explanation: domains characterised by standardised reporting (orthopaedic interventions and metabolic trials typically follow structured protocols with explicit randomisation and blinding procedures) show higher Macro-F1, while domains with heterogeneous or complex study designs (Infectious Disease, Reproductive & Sexual Health) present harder RoB characterisation challenges for models trained on general text. These observations motivate targeted future evaluation on larger domain-specific corpora.

3.7 Ground-Truth Uncertainty and Inter-Annotator Agreement

The benchmark treats published Cochrane annotations as the reference standard, but those annotations carry their own measurement error: independent Cochrane reviewer pairs achieve $\kappa = 0.24$–0.37 for non-D1 domains (Section 4). Figure 9 places model performance in this context by displaying the full 9×9 label-agreement matrix across Ground Truth and all eight models.

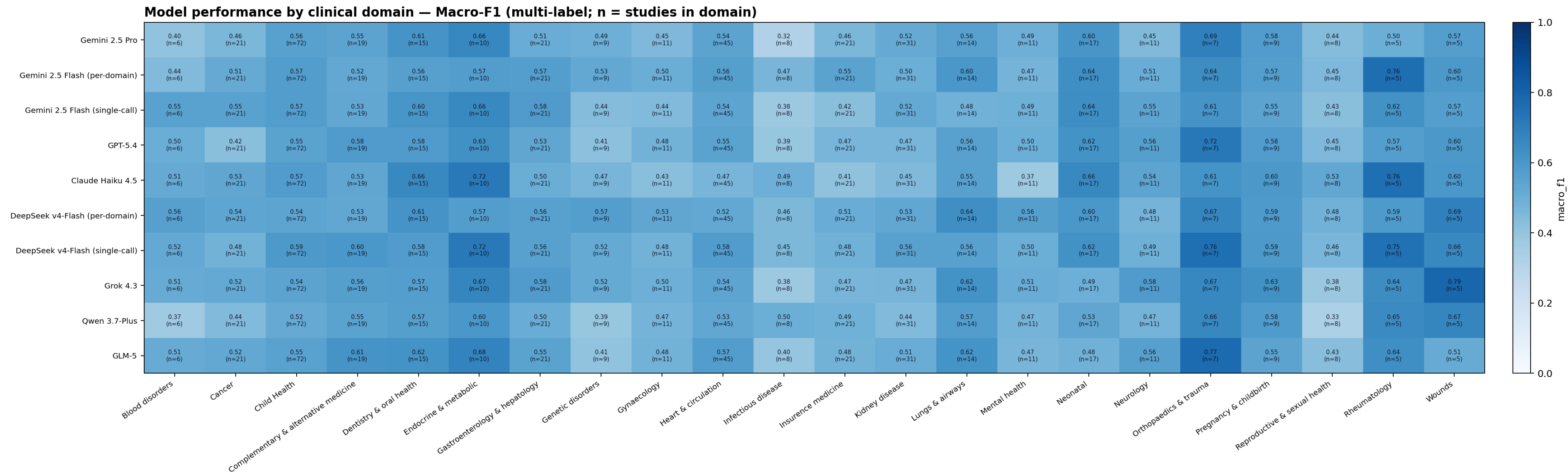

Figure 8: Macro-F1 by clinical domain and model (domains with ≥ 5 studies; n = number of studies in each domain, multi-label). Colour scale: 0–1.

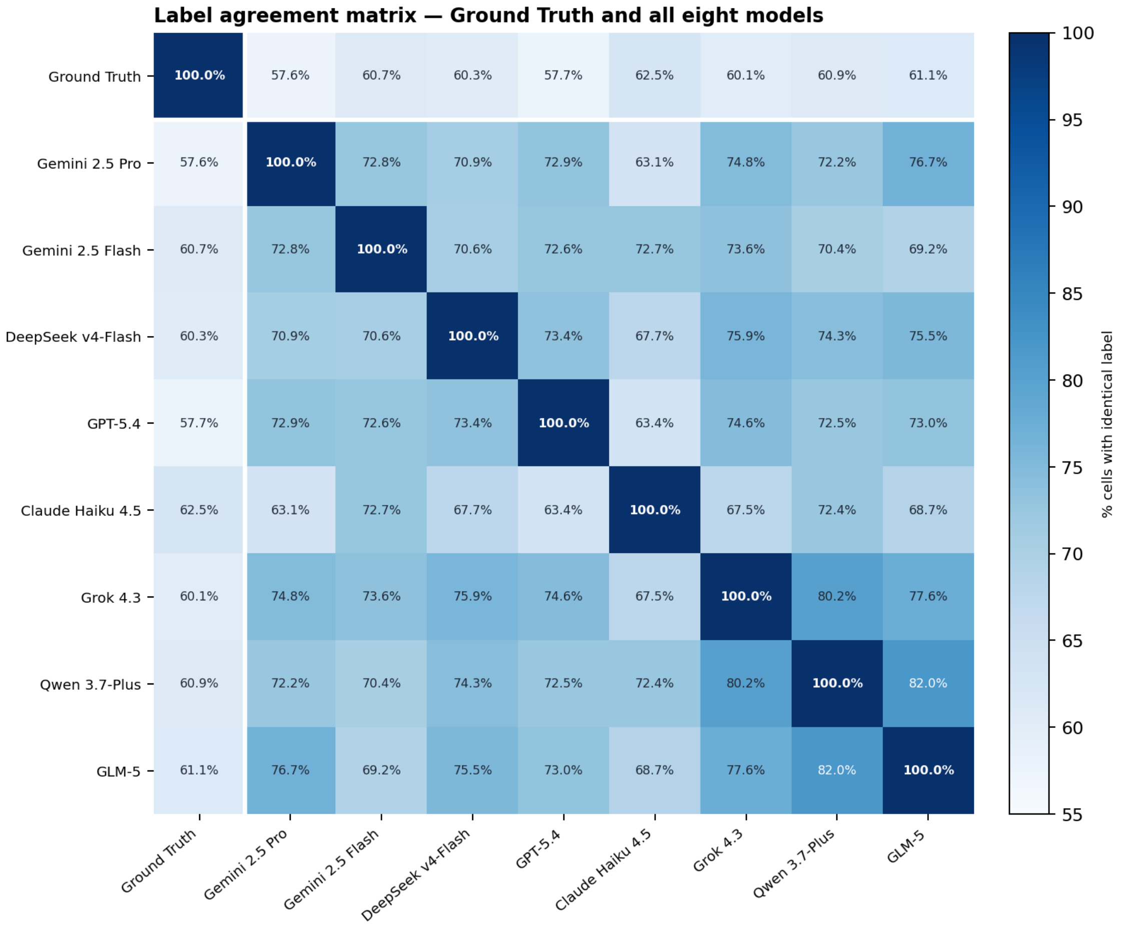

Figure 9: Label agreement matrix for Ground Truth and all eight models ($n = 1{,}589$ cells). Cell (i, j) = percentage of cells in which annotator i and annotator j assigned the *same* label, irrespective of correctness. The diagonal is 100% by definition. The white lines isolate the Ground Truth row and column. Colour scale: 55–100% (Blues).

The Ground Truth row (first row) reproduces the known model accuracies (57.6%–62.5%), representing agreement between each model and the Cochrane reference. All inter-model cells (the lower-right 8×8 block) exceed these values, ranging from 63.1% (Claude Haiku 4.5 and Gemini 2.5 Pro) to 82.0% (Qwen 3.7-Plus and GLM-5). These pairwise statistics are descriptive: no bootstrap confidence intervals were computed for them. Because cells from the same study share textual context and are correlated across domains, the effective sampling unit is the study ($n = 227$), not the cell ($n = 1{,}589$); each pairwise proportion therefore carries an approximate 95% uncertainty of ± 6 pp (SE $\approx \sqrt{p(1-p)/227}$, evaluated at $p \approx 0.70$). Differences smaller than this margin should not be over-interpreted; the patterns noted below reflect the overall structure of the matrix rather than fine-grained rankings between adjacent cells. This systematic gap—models agree with each other more than they agree with the ground truth—indicates that the eight models share systematic prediction tendencies that diverge from the Cochrane standard in coordinated ways. It does not imply that the models are collectively more reliable than the ground truth; rather, it reflects shared training priors that cause convergent errors.

The magnitude of the gap is interpretable in light of the known reference noise: if Cochrane reviewer pairs themselves agree at ≈60–79% (the $\kappa = 0.24$–0.37 range, converted to expected agreement under the observed class distribution), then the inter-model agreement range of 63–82% is broadly comparable to or modestly above human inter-rater reliability on this task. The ground-truth column therefore represents one reference annotator among many possible ones, and the model accuracy figures (57–63%) should be interpreted as lower bounds on the models' latent capability rather than definitive measurements of their performance against a noise-free standard.

3.8 Consensus Failures and Model Complementarity

Of the 1,589 study–domain cells, 226 (14.2%) were *consensus failures*: no model in the ensemble produced the correct judgement. These cells may reflect genuine textual ambiguity, inconsistency in the Cochrane annotation, or systematic limitations shared across all frontier models evaluated here. Consensus failures were concentrated in the most difficult domains: 50 (22.1%) occurred in D4 and 44 (19.5%) in D5—the two domains with the most balanced three-class distributions—corresponding to consensus failure rates of 22.0% and 19.4% in those domains, respectively; D2 had the lowest consensus failure rate (8.8%). Figure 10 shows the pairwise intersection matrix: cell (i, j) gives $P(A_i \cap A_j)$, the percentage of the 1,589 cells in which *both* model i and model j produced the correct judgement. The diagonal entries report each model's overall accuracy ($P(A_i)$).

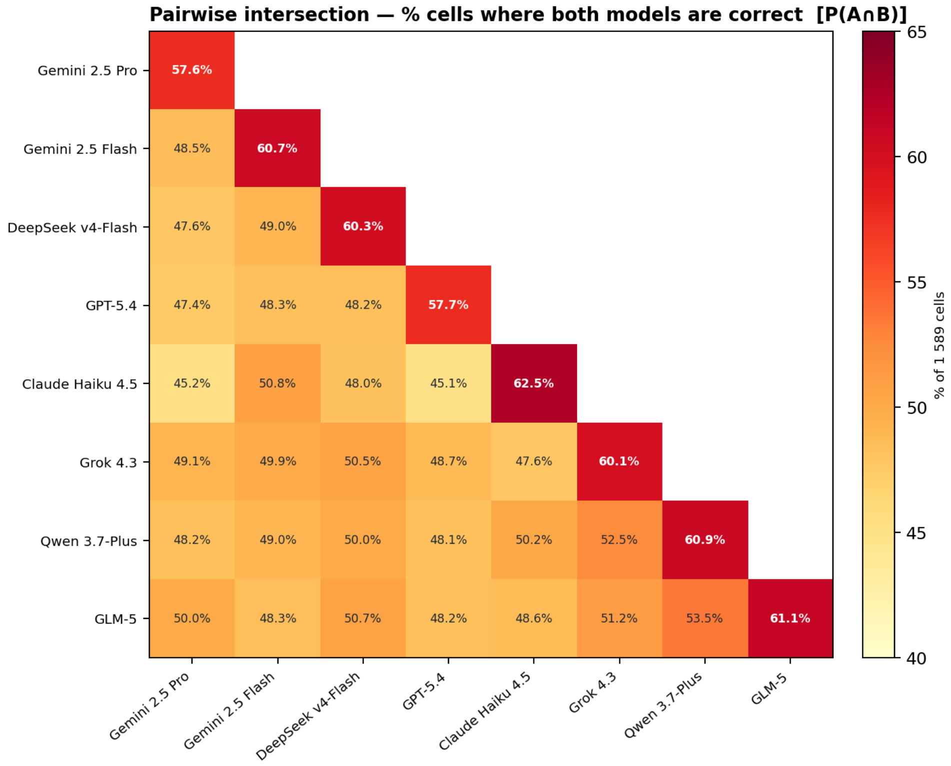

Figure 10: Pairwise intersection matrix. Cell $(i, j) = P(A_i \cap A_j)$: percentage of the 1,589 study–domain cells in which both models were correct. Diagonal entries (bold) = individual model accuracy. Colour scale: 40–65% (YlOrRd); deeper red = greater overlap in correct cells. Upper triangle omitted by symmetry.

Off-diagonal intersection values range from 45.1% (Claude Haiku 4.5 and GPT-5.4) to 53.5% (Qwen 3.7-Plus and GLM-5), all substantially below the respective diagonal entries (57.6%–62.5%). The gap between diagonal and off-diagonal reflects that no two models share all of their correct cells: a non-trivial fraction of each model's successes are unique to that model. Expressing overlap as the Jaccard similarity $J(A, B) = P(A \cap B) \,/\, P(A \cup B)$—the fraction of cells correct for *at least one* model that are correct for *both*—yields values between 60.0% and 78.1% across the 28 pairs. The highest Jaccard, Qwen 3.7-Plus and GLM-5 ($J = 78.1\%$, $P(A \cap B) = 53.5\%$), indicates that nearly four in five cells that either model answers correctly are answered correctly by both, confirming this as the most redundant pair. The lowest Jaccard, Claude Haiku 4.5 and GPT-5.4 ($J = 60.0\%$, $P(A \cap B) = 45.1\%$), indicates that two in five jointly-correct cells belong exclusively to one model—the highest non-redundancy in the ensemble and a direct predictor of the high complementarity entry visible in

Figure 11.

Figure 11 shows the full pairwise complementarity matrix: cell (i, j) gives the percentage of the 1,589 cells in which model i erred and model j was correct. Values range from 7.4% to 17.4%, with all off-diagonal entries exceeding 7%, confirming that no two models in the ensemble are redundant.

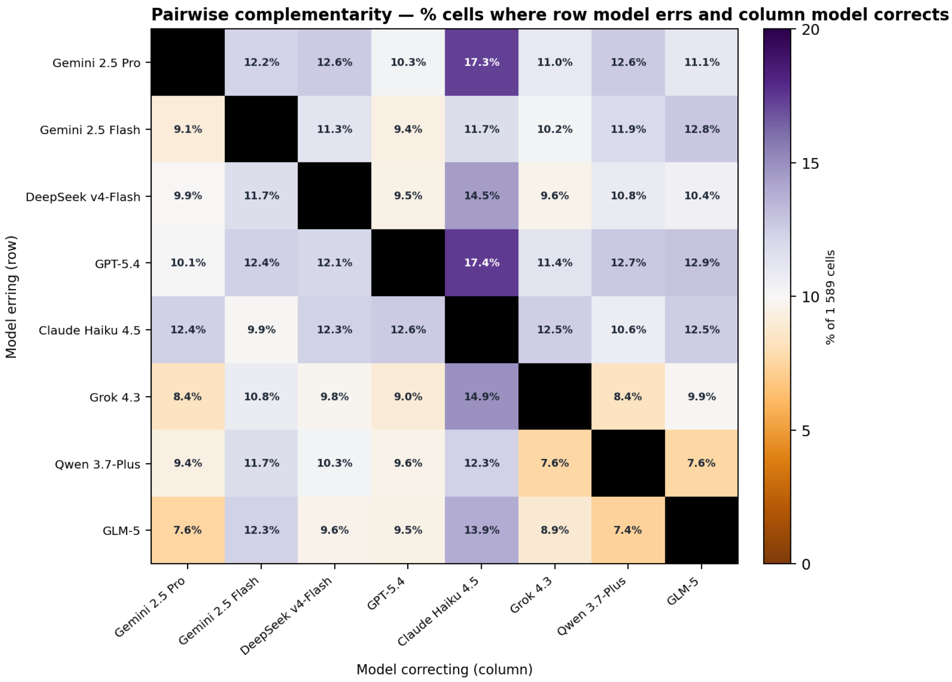

Figure 11: Pairwise complementarity matrix. Cell (i, j) = percentage of the 1,589 study–domain cells in which the row model erred and the column model was correct. Colour scale: 0–20% (PuOr); purple = high complementarity, orange = low complementarity; grey diagonal = undefined. Full numeric values are reported in Table SS6.

The most salient feature of Figure 11 is the Claude Haiku 4.5 column: it is the darkest column overall, meaning that when any other model errs, Haiku recovers those cells at the highest rate (12.3%–17.4%). This is the direct complement of its accuracy profile—Haiku leads on overall accuracy and AC1 by concentrating predictions on the majority class, which makes its correct cells less likely to be errors for other models, but also makes its errors structurally distinct from theirs. The two highest individual entries in the matrix—GPT-5.4 → Haiku (17.4%) and Gemini 2.5 Pro → Haiku (17.3%)—correspond to the two models with the lowest overall accuracy; their larger error sets overlap least with Haiku's. The same structural distinctiveness is visible in the agreement matrix (Figure 9): Haiku has the lowest mean inter-model agreement across the ensemble (67.9%, versus 71.7%–74.9% for all other models), driven by consistently low agreement with Gemini 2.5 Pro and GPT-5.4 (63–64%)—the same dyads that top the complementarity column. High complementarity and low overall agreement are two sides of the same coin: because Haiku assigns labels differently from the ensemble cluster, it agrees with other models less often irrespective of correctness, but those disagreements are disproportionately cases where Haiku is right and the other model is wrong. At the opposite extreme, the Qwen 3.7-Plus ↔ GLM-5 pair shows the lowest bidirectional complementarity (7.4% and 7.6%), indicating the most overlapping error profiles of any pair in the ensemble. Grok 4.3 also shows low complementarity as a *column* model for the Qwen and GLM-5 rows (7.6% and 8.9%), suggesting that these three models share a similar error structure that is not well covered by the others. This pattern is reinforced by the agreement matrix (Figure 9): the three highest pairwise inter-model agreements—Qwen 3.7-Plus / GLM-5 (82.0%), GLM-5 / Grok 4.3 (80.2%), and Qwen 3.7-Plus / Grok 4.3 (77.6%)—are all concentrated within this trio; the trio's mean agreement (79.9%) exceeds the mean of cross-trio pairs by 7.2 pp. Taken together, the matrix indicates that Claude Haiku 4.5, Gemini 2.5 Pro, and GPT-5.4 form the most complementary triad: their pairwise entries are uniformly above 10%, and combining any two of them recovers more than a fifth of all cells that either individually misses.

3.9 Inference Mode Comparison

Table 3 compares per-domain and single-call inference modes for the two models evaluated under both conditions.

The overall picture from this two-model pilot comparison (Table 3) is one of small differences with wide uncertainty: of 12 Δ values across both models and six metrics, ten have 95% CIs that include zero. The two whose CIs exclude zero are a drop in Unclear Risk F1 for Gemini 2.5 Flash (-0.044; CI $[-0.079, -0.009]$, †) and a gain in Low Risk F1 for DeepSeek v4-Flash ($+0.026$; $[+0.004, +0.047]$, †);

Table 3: Per-domain vs. single-call inference mode comparison for Gemini 2.5 Flash and DeepSeek v4-Flash ($n = 227$ studies; 1,589 cells each). F1 scores are computed globally across all cells (identical methodology to Table SS1); per-domain values reproduce the corresponding entries in Table SS1. Δ: single-call minus per-domain; 95% paired bootstrap CI (study-clustered, 5,000 iterations) shown in brackets. †: CI excludes zero (unadjusted for multiplicity; all analyses exploratory).

Model	Accuracy (%)	Macro-F1	F1-LR	F1-HR	F1-UR	Gwet's AC1
Gemini 2.5 Flash						
Per-domain	60.7	0.558	0.718	0.474	0.484	0.437
Single-call	59.8	0.548	0.715	0.490	0.440	0.425
Δ	−0.9	−0.010	−0.003	+0.016	−0.044†	−0.012
	[−2.7, +1.0]	[−0.030, +0.011]	[−0.019, +0.014]	[−0.013, +0.051]	[−0.079, −0.009]	[−0.038, +0.015]
DeepSeek v4-Flash						
Per-domain	60.3	0.558	0.704	0.480	0.489	0.433
Single-call	62.6	0.576	0.730	0.475	0.523	0.466
Δ	+2.3	+0.018	+0.026†	−0.005	+0.034	+0.033
	[−0.1, +4.7]	[−0.007, +0.044]	[+0.004, +0.047]	[−0.049, +0.041]	[−0.003, +0.072]	[−0.001, +0.066]

both are class-specific and do not translate into differences in accuracy, Macro-F1, or AC1, all of which remain within their respective uncertainty intervals. The accuracy differences (−0.9 pp for Flash, +2.3 pp for DeepSeek), while pointing in opposite directions, are each compatible with zero change given the bootstrap CIs ($[-2.7, +1.0]$ and $[-0.1, +4.7]$ respectively).

Domain-level results reinforce this picture. D7 (Other Bias), the domain for which joint assessment was hypothesised to confer the greatest benefit through shared contextual reasoning, showed negligible differences under both modes (±1 pp)— the most direct evidence that collapsing seven domain assessments into a single call does not meaningfully improve the handling of the hardest domain. The opposing D3 pattern (Flash: −0.4 pp; DeepSeek: −3.6 pp) and the concentration of DeepSeek gains in D5 (+10.2 pp) and D2 (+4.4 pp) suggest that whatever domain-level shifts occur are idiosyncratic to each model rather than reflecting a systematic benefit of joint assessment.

Single-call outputs were systematically shorter (DeepSeek: 400→214 chars; Gemini 2.5 Flash: 530→354 chars), with a corresponding decrease in hedging expressions (e.g., “unclear”, “not reported”, “cannot determine”) in free-text justifications, consistent with a more compressed generation regime under the single-call prompt. The practical implications of this output compression appear limited and modeldependent: it did not translate into a consistent performance benefit or detriment across either model at the aggregate level.

4 Conclusions

Eight state-of-the-art LLMs assessed RoB 1.0 for 227 randomised controlled trials in a zero-shot, single-agent setting. Overall accuracy ranged narrowly from 57.6% to 62.5%, but Macro-F1 and Gwet's AC1 revealed meaningful differences in how models handled the three nominal classes. Models with the highest accuracy achieved it partly by concentrating predictions on majority-class labels, a strategy that inflates accuracy whilst suppressing detection of High Risk and Unclear Risk judgements—the very categories most consequential for evidence appraisal. Macro-F1 and Gwet's AC1 are therefore more informative primary metrics than accuracy for this task.

Unclear Risk was the most consistently under-detected class across all models and domains, with recall below 0.20 in three of seven domains. This systematic failure reflects the epistemic character of the category—it requires the model to recognise that key methodological information is absent, a more complex inference than identifying a positive or negative methodological signal in the text.

Domain-level and stratification analyses revealed structured heterogeneity in performance. Sequence generation (D1) and allocation concealment (D2) were reliably assessed across all models, whereas other sources of bias (D7) showed the greatest inter-model accuracy spread and blinding of outcome assessors (D4) had the highest consensus failure rate (22.0%—no model correct in 50 of 227 cells). Clinical domains with clear, domain-specific methodological criteria (Orthopaedics & Trauma, Endocrine & Metabolic) supported higher Macro-F1 than domains with more heterogeneous study designs (Infectious Disease, Reproductive & Sexual Health).

Consensus failures—cells where no model was correct—accounted for 14.2% (226 of 1,589 cells) and were concentrated in D4 and D5, the two most ambiguous domains. These cases identify the boundaries of current frontier model capability under this evaluation design and may reflect genuine annotation uncertainty as much as model failure. The remaining 85.8% of cells had at least one correct model, yet pairwise complementarity analysis showed that 7.4%–17.4% of cells are correctable by substituting any one model for another, confirming that the eight models make substantially non-redundant errors.

The comparison between per-domain and single-call inference modes (Section 3.9) produced small differences that do not collectively support a performance advantage for either strategy. Ten of twelve Δ values across both models and six metrics had 95% CIs spanning zero; the two exceptions were class-specific effects of limited practical magnitude (Unclear Risk F1 for Flash: -0.044; Low Risk F1 for DeepSeek: $+0.026$) that did not propagate to aggregate metrics. Most notably, D7 (Other Bias)—the domain for which joint contextual assessment was hypothesised to be most beneficial—showed negligible differences under both modes (± 1 pp), indicating that its difficulty is intrinsic to the vagueness of the criterion rather than a consequence of fragmented inference calls. Taken together, the preliminary evidence from this two-model comparison does not identify a meaningful performance rationale for preferring single-call over per-domain assessment; the primary practical distinction between the two modes is computational (single-call reduces inference calls by 7×) rather than one of accuracy.

Reframing High Risk detection as a binary classification task (High Risk vs. not High Risk) highlights the clinical stakes of the leniency bias documented above. Under this framing, the best-performing models for High Risk recall detected fewer than two thirds of High Risk cells (F1-HR: 0.455–0.480 for the top-ranked models), meaning roughly one in three cells that should trigger reviewer scrutiny passed undetected. Because the cost of a false negative—a biased study classified as acceptable and included unquestioned in a meta-analysis—substantially exceeds the cost of a false positive, tools intended to assist real-world screening workflows would need to prioritise sensitivity over overall accuracy, a trade-off not currently optimised by the zero-shot prompt evaluated here.

The cell-level concordance partition offers a practical framework for directing qualitative review effort. Cells in which all models converge on the correct answer require no additional attention. Cells in which all models converge on the *same wrong* answer are the highest-priority targets for prompt refinement or annotation review: unanimity of error across architecturally diverse models makes model-idiosyncratic explanation implausible, pointing instead to shared training priors, prompt ambiguity, or genuine ground-truth noise. Cells in which a single model is the sole outlier—either the only model correct or the only model incorrect—index model-specific capacity gaps independent of task difficulty. Targeting qualitative analysis at consensus errors before individual outliers maximises diagnostic yield per unit of human effort.

The domain and stratification analyses converge on a specific, actionable direction for prompt improvement: a blinding rule sensitive to intervention type. Models would benefit from explicit guidance distinguishing trials in which blinding is physically feasible (pharmacological interventions with matched placebos) from those in which it is not (surgical procedures, physiotherapy, medical devices), with the latter receiving a conservative default that does not presuppose the availability of a blinding mechanism. This single rule targets D4—the domain with the most systematic and directionally consistent errors in the benchmark—and has direct implications for the High Risk recall deficit identified above.

Several limitations should be noted. First, the ground-truth annotations derive from published Cochrane review reports and were not independently re-annotated for this study. The Cochrane annotations represent the consensus output of a structured multi-reviewer process (independent assessment by at least two trained reviewers, with disagreements resolved by discussion or arbitration), which is the reference standard used operationally by the field and the most appropriate benchmark for evaluating a tool intended to assist real-world systematic review workflows. Quantifying internal reliability within this corpus would require re-annotating a subsample with independent trained reviewers; however, agreement between Cochrane-trained reviewers and non-specialist re-annotators would conflate task difficulty with expertise asymmetry—the same confound that limits the interpretability of comparisons between Cochrane and external reviewers [5]. The relevant population-level IRR estimate is therefore taken from Hartling et al. [1], who measured agreement between formally trained reviewer pairs applying the same protocol: $\kappa = 0.79$ for D1 but only $\kappa = 0.24$–0.37 for remaining domains, falling to $\kappa = 0.05$–0.09 between independent consensus pairs; independent Cochrane teams show similarly variable agreement [4]. These figures establish that the reference standard itself carries substantial noise, placing a practical ceiling on any automated system's agreement with published annotations that is independent of model quality. Converting the domain-level κ values for the non-D1 domains (0.24–0.37) to expected accuracy using the observed pooled class distribution (LR 53.5%, HR 17.4%, UR 29.1%) yields approximately 58–62%—a range that coincides closely with the model accuracy range observed in this study (57.6%–62.5%), suggesting that the evaluated LLMs perform at a level broadly comparable to a second independent human reviewer across most domains. Second, each model was queried once per study using provider-default generation parameters. This reflects the intended evaluation scope: zero-shot performance as models are available to practitioners, without artificial parameter tuning. Manipulating generation temperature would introduce heterogeneous conditions across model families, and strategies such as majority voting over repeated draws represent post-hoc optimisation outside the zero-shot benchmark design. As a consequence, differences between models separated by fewer than five percentage points should not be interpreted as definitive rank orderings in the absence of confidence intervals; these are discussed as directional trends. Third, all models were accessed in June 2026 via the OpenRouter API using the identifier strings reported in Table 2; providers may silently update the weights associated with a fixed identifier, so exact reproduction requires accessing the same model strings within the same inference window. Results should therefore be interpreted as reflecting the checkpoint state active in June 2026, not any future revision of the same identifier. Fourth, the single-agent zero-shot design examines one specific slice of the capability space; few-shot prompting, retrieval augmentation, fine-tuning, and multi-agent architectures were not evaluated and may yield substantially different results. Fifth, training-data contamination cannot be excluded, but several structural features of the evaluation design limit its plausibility as a primary driver of results. Memorisation-based performance would require a model to (i) have seen the particular one of the 127 distinct Cochrane 2024 reviews from which each study was drawn, (ii) correctly identify that the input Markdown document corresponds to a specific entry in that review's RoB table, and (iii) reproduce the reviewer's label rather than reason from the text. The Cochrane reviews sampled here were published in 2024; most included RCTs appear in paywalled journals, making full-text ingestion at training time less likely for the majority of papers. Moreover, the empirical pattern is inconsistent with memorisation at scale: the best individual model (Claude Haiku 4.5) achieved a perfect 7/7 score on only 12 of 227 studies (5.3%), and even across the full ensemble of eight models all seven domains were correct in only 40 studies (17.6%). Both figures are consistent with the rates expected under independent domain judgements at the observed per-domain accuracy ($0.60^7 \approx 2.8\%$; inflated modestly by the low intraclass correlation of $\hat{\rho} = 0.044$ to roughly 4–5% per model; see Supporting Information S7 for derivation), leaving no systematic excess attributable to retrieval. Targeted contamination probes (e.g. membership inference or verbatim reproduction tests) were not conducted, as the opaque training corpora of the evaluated models preclude rigorous analysis; this remains a caveat for interpretation. Sixth, the Cochrane corpus used here comprises better-reported trials than the broader trial literature, so these performance estimates may be optimistic relative to unselected primary research.

Acknowledgements

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Data and Code Availability

The analysis code is publicly available at `https://github.com/ACS-USP/systematic-reviews`.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

S1 Prompt Used for Single-Agent RoB Assessment

The following prompt was used without modification across all eight models and all 227 studies. Each study was assessed in seven sequential inference calls—one per RoB domain—with the full Markdown text of the study report included in every call as the user message, followed by the domain identifier (e.g. `Domain: D1`). The prompt text below was supplied as the system message.

```
Goal: Systematically assess the risk of bias in a research article using
Cochrane RoB 1.0. Evaluate bias domain, assign a judgment (Low, High,
or Unclear Risk), and justify strictly from the article evidence.

---

Preliminary Considerations

Before assessing, identify:
1. Intervention Type: Pharmacological | Surgical/Procedural |
   Non-pharmacological | Device
2. Primary Outcome Type: Objective (mortality, lab values) |
   Subjective (pain, QoL)
3. Comparator: Placebo | Active comparator | Usual care | No intervention

Critical Context: Blinding expectations vary by intervention type.
For pharmacological interventions, lack of placebo is a serious flaw.
For surgical/non-pharmacological interventions where blinding is impossible,
focus on compensatory measures: blinding of outcome assessors and objective
endpoints.

---

Bias Domains
```

1. Random Sequence Generation (Selection Bias)

Signalling Question: Was a truly random method used?

Low Risk: Random method adequately described (computer-generated, random tables, coin toss, dice, shuffled envelopes, minimization with random element)

High Risk: Non-random method (alternation, date of birth, hospital record number, clinician judgment)

Unclear Risk: States "randomized" but no method details provided

2. Allocation Concealment (Selection Bias)

Signalling Question: Was the allocation sequence concealed from those enrolling participants until assignment?

Low Risk: Adequate concealment (central/telephone randomization, sequentially numbered opaque sealed envelopes, numbered drug containers, independent pharmacy)

High Risk: Allocation not concealed or predictable (open list, non-opaque envelopes, alternation)

Unclear Risk: Insufficient information to judge

Note: Allocation concealment (before randomization) is different from blinding (after randomization).

3. Blinding of Participants and Personnel (Performance Bias)

Signalling Questions: Were participants/personnel aware of assignments? If yes, could this affect outcomes?

Pharmacological: Double-blinding expected; no placebo implies higher risk for subjective outcomes.

Surgical/Non-pharmacological: Blinding may be impossible and is not automatically High Risk; check compensatory safeguards.

Objective outcomes: Less susceptible to performance bias.

Low Risk: Adequate blinding, OR blinding impossible with compensatory measures, OR outcome is objective/hard endpoint

High Risk: No blinding + subjective outcome + no compensatory measures

Unclear Risk: Insufficient blinding information

4. Blinding of Outcome Assessors (Detection Bias)

Signalling Questions: Were assessors blinded? Is outcome objective or subjective?

Low Risk: Assessors blinded, OR objective outcome unlikely to be influenced by assignment knowledge

High Risk: Assessors not blinded + subjective outcome requiring interpretation

Unclear Risk: Insufficient information

Note: For patient-reported outcomes, participant acts as assessor.

5. Incomplete Outcome Data (Attrition Bias)

Signalling Questions: Are data available for most participants? Are missing-data reasons balanced/unrelated to outcome?

Low Risk: Minimal missingness (<5%), OR balanced missingness with unrelated reasons, OR valid ITT/imputation

High Risk: High missingness (>10-20%), OR imbalance between groups, OR outcome-related missingness, OR per-protocol exclusions

Unclear Risk: Missingness not adequately reported

6. Selective Reporting (Reporting Bias)

Signalling Questions: Is protocol/registry available? Were pre-specified outcomes reported?

Low Risk: All pre-specified outcomes reported, OR protocol/registry aligns with publication

High Risk: Selective non-reporting (methods outcomes missing in results, changed primary outcome, significance-driven reporting)

```
   Unclear Risk: No protocol/registry and insufficient information

7. Other Bias
   Signalling Questions: Are there other bias sources (early stopping,
   baseline imbalance, conflicts, contamination)?
   Low Risk:    No relevant additional bias detected
   High Risk:   Important additional bias present
   Unclear Risk: Insufficient information

---

Key Principles
1. Evidence-based only: Base judgments on explicit information. Do not
   assume. If evidence is missing, choose "Unclear Risk".
2. Cite evidence: Reference specific text or note absence of information.
3. Context matters: Apply blinding expectations appropriate to intervention
   type.
4. Impossibility vs Negligence: Not blinding when impossible differs from
   not blinding when feasible.
5. Outcome sensitivity: Subjective outcomes require stricter blinding than
   objective ones.

---

Return ONLY valid JSON with:
{
  "judgement": "Low Risk" | "High Risk" | "Unclear Risk",
  "justification": "short justification"
}
```

Table S1: Overall performance on RoB 1.0 assessment ($n = 227$ studies; 1,589 cells). Models ranked by accuracy. F1-LR, F1-HR, F1-UR: F1 for Low Risk, High Risk, and Unclear Risk computed globally across all study–domain cells. Best value in each column shown in bold. 95% confidence intervals (study-clustered bootstrap, 5,000 iterations, resampling 227 studies with replacement) are shown in *brackets* below each point estimate; all analyses are exploratory (see Methods, Statistical Inference).

Model	Accuracy (%)	Macro-F1	F1-LR	F1-HR	F1-UR	Gwet's AC1
Claude Haiku 4.5	**62.5**	0.541	**0.743**	0.469	0.413	**0.485**
	[59.8, 65.2]	[0.511, 0.569]	[0.718, 0.767]	[0.412, 0.521]	[0.361, 0.463]	[0.443, 0.526]
GLM-5	61.1	0.557	0.709	0.441	**0.520**	0.449
	[58.7, 63.5]	[0.529, 0.582]	[0.682, 0.735]	[0.385, 0.494]	[0.479, 0.561]	[0.411, 0.485]
Qwen 3.7-Plus	60.9	0.525	0.718	0.356	0.503	0.457
	[58.3, 63.4]	[0.496, 0.552]	[0.691, 0.743]	[0.293, 0.413]	[0.459, 0.543]	[0.417, 0.495]
Gemini 2.5 Flash	60.7	**0.558**	0.718	**0.474**	0.484	0.437
	[58.2, 63.1]	[0.533, 0.582]	[0.690, 0.744]	[0.424, 0.516]	[0.443, 0.526]	[0.399, 0.475]
DeepSeek v4-Flash	60.3	**0.558**	0.704	**0.480**	0.489	0.433
	[57.7, 62.9]	[0.531, 0.585]	[0.676, 0.730]	[0.429, 0.530]	[0.447, 0.530]	[0.394, 0.472]
Grok 4.3	60.1	0.549	0.696	0.432	0.518	0.433
	[57.8, 62.5]	[0.523, 0.574]	[0.670, 0.722]	[0.376, 0.486]	[0.478, 0.556]	[0.397, 0.471]
GPT-5.4	57.7	0.546	0.668	0.464	0.506	0.387
	[55.1, 60.3]	[0.519, 0.572]	[0.636, 0.698]	[0.410, 0.511]	[0.468, 0.543]	[0.349, 0.426]
Gemini 2.5 Pro	57.6	0.543	0.674	0.455	0.502	0.383
	[55.1, 60.0]	[0.517, 0.569]	[0.643, 0.702]	[0.407, 0.501]	[0.458, 0.542]	[0.344, 0.420]

S2 Overall Model Performance

S3 Per-Domain Performance by Model

Tables S2 and S3 report accuracy (%) and Macro-F1 for each model in each of the seven RoB domains.

S4 Stratification by Intervention Type

Figure S1 shows the distribution of intervention types within each clinical domain in the 227-study corpus. Because each study can be tagged with multiple clinical domains, the counts across domains sum to more than 227; a study tagged with two domains contributes its intervention type to both. The distribution is markedly non-uniform: Pregnancy & childbirth and Complementary & alternative medicine are almost exclusively pharmacological, Effective practice & health systems and Mental health are predominantly non-pharmacological, and Orthopaedics & trauma contains no pharmacological studies. Table S4 reports Macro-F1 for each model within each intervention type. Figure S2 shows the same data as a heatmap.

Table S2: Per-domain accuracy (%) for all eight models ($n = 227$ cells per domain). Best value per domain in bold.

Model	D1	D2	D3	D4	D5	D6	D7
Claude Haiku 4.5	73.6	74.0	67.4	51.5	**64.3**	57.7	48.9
GLM-5	76.7	74.0	55.9	53.7	62.1	55.9	49.3
Qwen 3.7-Plus	**77.5**	74.9	49.8	52.0	59.9	56.4	**55.5**
Gemini 2.5 Flash	72.2	74.0	**72.2**	51.5	60.4	**62.6**	31.7
DeepSeek v4-Flash	74.0	70.0	67.0	**57.7**	54.6	55.9	42.7
Grok 4.3	75.8	70.0	59.5	54.6	58.6	60.8	41.4
GPT-5.4	75.8	70.0	64.3	52.0	51.5	55.9	34.4
Gemini 2.5 Pro	76.7	**74.4**	62.6	55.1	55.1	55.1	24.2

Table S3: Per-domain Macro-F1 for all eight models ($n = 227$ cells per domain). Best value per domain in bold.

Model	D1	D2	D3	D4	D5	D6	D7
Claude Haiku 4.5	0.572	0.618	0.527	0.425	0.394	0.387	0.269
GLM-5	0.632	0.657	0.462	0.479	**0.514**	0.443	0.345
Qwen 3.7-Plus	**0.638**	0.672	0.428	0.447	0.439	0.388	0.288
Gemini 2.5 Flash	0.549	0.663	**0.626**	0.411	0.412	0.438	0.280
DeepSeek v4-Flash	0.602	0.610	0.583	**0.521**	0.419	0.424	**0.347**
Grok 4.3	0.618	0.668	0.524	0.492	0.423	**0.449**	0.315
GPT-5.4	0.625	0.635	0.515	0.469	0.419	0.397	0.317
Gemini 2.5 Pro	0.621	**0.671**	0.481	0.478	0.413	0.475	0.195

Clinical domain × intervention type

Clinical domain	Total n	Drug / medication	Non-pharm. intervention	Procedure	Medical device
Child Health	**72**	29	27	8	8
Heart & circulation	**45**	16	22	4	3
Kidney disease	**31**	16	8	5	2
Gastroenterology & hepatology	**21**	8	4	7	2
Insurence medicine	**21**	10	5	6	0
Cancer	**21**	14	1	6	0
Complementary & alternative medicine	**19**	14	4	1	0
Neonatal	**17**	3	9	1	4
Dentistry & oral health	**15**	6	3	3	3
Lungs & airways	**14**	8	2	0	4
Gynaecology	**11**	5	2	4	0
Neurology	**11**	8	2	0	1
Mental health	**11**	3	8	0	0
Endocrine & metabolic	**10**	6	3	1	0
Genetic disorders	**9**	7	2	0	0
Pregnancy & childbirth	**9**	9	0	0	0
Infectious disease	**8**	6	1	0	1
Reproductive & sexual health	**8**	3	4	1	0
Orthopaedics & trauma	**7**	1	4	2	0
Blood disorders	**6**	4	0	2	0
Rheumatology	**5**	4	1	0	0
Wounds	**5**	0	2	1	2
Effective practice & health systems	**4**	0	4	0	0
Skin disorders	**3**	0	2	1	0

Figure S1: Distribution of intervention types across clinical domains ($n = 227$ studies; studies with multiple clinical domain tags are counted in each relevant domain). Cell colour encodes the fraction of that domain's studies belonging to each intervention type (pastel red = low proportion, pastel blue = high proportion); grey cells indicate zero studies. Domains are sorted by total study count (descending).

S5 Stratification by Clinical Domain

Table S5 reports mean Macro-F1 (averaged across all eight models) for all 24 Cochrane Review Group categories, ordered from highest to lowest mean F1. Domains with fewer than five studies are included for completeness but should be interpreted with caution.

Table S4: Macro-F1 stratified by intervention type. Numbers in parentheses are study counts (n_{studies}). Best value per row in bold. Subgroup estimates carry substantial uncertainty given the small group sizes; these analyses are descriptive only.

Model	Drug / medication ($n = 104$)	Non-pharma-cological ($n = 75$)	Procedure ($n = 30$)	Medical device ($n = 18$)
Claude Haiku 4.5	0.548	0.530	0.494	**0.627**
GLM-5	**0.559**	**0.590**	0.514	0.471
Qwen 3.7-Plus	**0.511**	0.567	0.474	0.445
Gemini 2.5 Flash	0.540	0.574	0.529	**0.596**
DeepSeek v4-Flash	0.548	**0.582**	**0.520**	0.543
Grok 4.3	**0.552**	0.561	0.519	0.531
GPT-5.4	0.515	**0.595**	**0.530**	0.502
Gemini 2.5 Pro	0.533	0.575	0.498	0.534

S6 Pairwise Model Complementarity

Table S6 reports pairwise complementarity across the eight models. Each cell (i, j) gives the percentage of the 1,589 study–domain cells in which model i produced an incorrect judgement and model j produced the correct one. High values indicate that model j recovers errors that model i misses, i.e. that j adds non-redundant information over i. The matrix is asymmetric: cell $(i, j) \neq$ cell (j, i) because the two models may have different overall accuracy.

S7 Confidence Interval Method Comparison

All confidence intervals reported in the main text were computed using study-clustered bootstrap (5,000 iterations, resampling the 227 studies with replacement). This section documents the comparison between bootstrap and two Wilson score interval variants for accuracy, and explains why the bootstrap approach was adopted for all metrics.

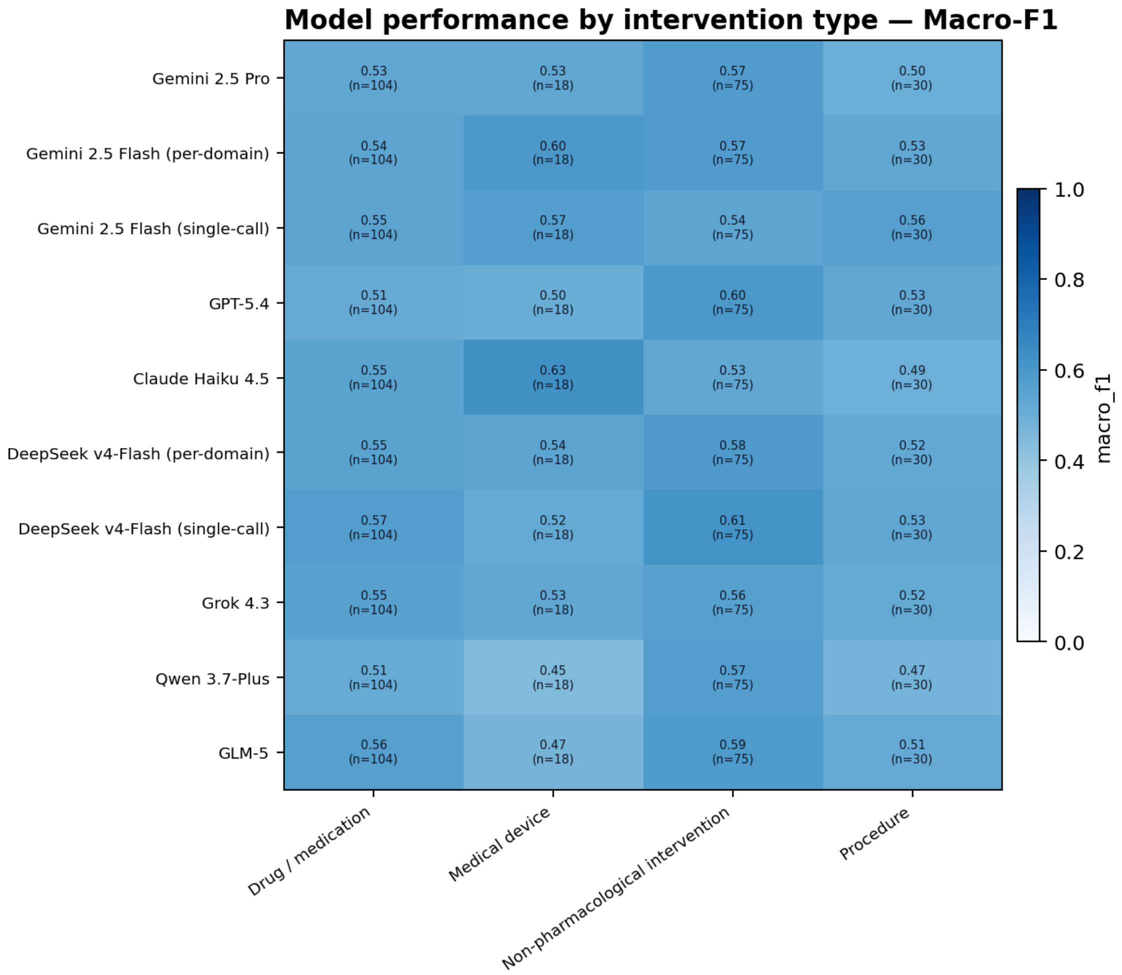

Figure S2: Macro-F1 heatmap by intervention type and model. Colour scale: 0–1.

Wilson score interval

For a proportion p observed over n independent Bernoulli trials, the Wilson 95% CI is

$$\left(\frac{np + \frac{z^2}{2}}{n + z^2} \pm \frac{z\sqrt{np(1-p) + \frac{z^2}{4}}}{n + z^2} \right), \tag{S1}$$

with $z = 1.96$. Applied to accuracy ($p \approx 0.60$), two natural choices for n exist: $n = 1{,}589$ (all annotated cells, assuming independence) or $n = 227$ (studies as the primary sampling unit).

Intraclass correlation and effective sample size

The 1,589 cells are not independent: seven domain judgements are drawn from the same study. Pearson's one-way intraclass correlation coefficient (ICC) for the

Table S5: Mean Macro-F1 across all eight models, stratified by clinical domain (24 Cochrane Review Groups, multi-label; study counts sum to more than 227). Ordered by descending mean F1. Estimates for smaller groups carry substantial uncertainty; these analyses are descriptive only.

Clinical domain	n_{studies}	Mean Macro-F1
Orthopaedics & trauma	7	0.680
Endocrine & metabolic	10	0.638
Rheumatology	5	0.637
Wounds	5	0.628
Dentistry & oral health	15	0.597
Lungs & airways	14	0.589
Pregnancy & childbirth	9	0.583
Neonatal	17	0.578
Complementary & alternative medicine	19	0.555
Child health	72	0.550
Gastroenterology & hepatology	21	0.538
Heart & circulation	45	0.536
Neurology	11	0.519
Cancer	21	0.494
Effective practice & health systems	4	0.487
Kidney disease	31	0.484
Mental health	11	0.481
Gynaecology	11	0.480
Insurance medicine	21	0.477
Blood disorders	6	0.476
Genetic disorders	9	0.473
Skin disorders	3	0.448
Reproductive & sexual health	8	0.437
Infectious disease	8	0.425

correct/wrong binary indicator across the seven domains within each study was estimated at $\hat{\rho} = 0.044$ (computed for Claude Haiku 4.5 as a representative model). The low ICC indicates that whether a model correctly judges one domain bears little information about its judgement on another domain of the same study, which is consistent with the domain-specific nature of each RoB criterion.

Using the Kish (1965) design-effect formula, the effective sample size is

$$n_{\text{eff}} = \frac{n_{\text{cells}}}{1 + (k-1)\hat{\rho}} = \frac{1{,}589}{1 + 6 \times 0.044} \approx 1{,}257, \tag{S2}$$

where $k = 7$ domains per study.

Table S6: Pairwise complementarity matrix. Cell (i, j) = % of 1,589 cells where row model i is incorrect and column model j is correct. Row model abbreviations: Haiku = Claude Haiku 4.5; GLM-5 = GLM-5; Qwen = Qwen 3.7-Plus; Flash = Gemini 2.5 Flash; DSv4 = DeepSeek v4-Flash; Grok = Grok 4.3; GPT = GPT-5.4; Pro = Gemini 2.5 Pro.

	Haiku	**GLM-5**	**Qwen**	**Flash**	**DSv4**	**Grok**	**GPT**	**Pro**
Haiku	—	12.5	10.6	9.9	12.3	12.5	12.6	12.4
GLM-5	13.9	—	7.4	12.3	9.6	8.9	9.5	7.6
Qwen	12.3	7.6	—	11.7	10.3	7.6	9.6	9.4
Flash	11.7	12.8	11.9	—	11.3	10.2	9.4	9.1
DSv4	14.5	10.4	10.8	11.7	—	9.6	9.5	9.9
Grok	14.9	9.9	8.4	10.8	9.8	—	9.0	8.4
GPT	17.4	12.9	12.7	12.4	12.1	11.4	—	10.1
Pro	17.3	11.1	12.6	12.2	12.6	11.0	10.3	—

Numerical comparison

Table S7 reports accuracy CIs for five models under three methods.

Table S7: Comparison of 95% confidence interval methods for overall accuracy. Width in percentage points (pp). Bootstrap: study-clustered, 5,000 iterations. Wilson n_{cells}: Equation (S1) with $n = 1{,}589$. Wilson n_{eff}: Equation (S1) with $n = 1{,}257$ (Kish formula). Wilson n_{studies}: Equation (S1) with $n = 227$.

Model	Accuracy	Wilson n_{cells} (width)	Wilson n_{eff} (width)	Bootstrap (width)
Claude Haiku 4.5	62.5%	[60.1, 64.8] (4.8)	[59.8, 65.1] (5.3)	[59.8, 65.2] (5.4)
Gemini 2.5 Flash	60.7%	[58.2, 63.0] (4.8)	[57.9, 63.4] (5.4)	[58.2, 63.1] (4.9)
DeepSeek v4-Flash	60.3%	[57.9, 62.7] (4.8)	[57.6, 63.0] (5.4)	[57.7, 62.9] (5.2)
GPT-5.4	57.7%	[55.3, 60.1] (4.9)	[54.9, 60.4] (5.4)	[55.1, 60.3] (5.2)
Gemini 2.5 Pro	57.6%	[55.1, 60.0] (4.9)	[54.8, 60.3] (5.4)	[55.1, 60.0] (4.9)
Wilson $n_{\text{studies}} = 227$		widths ≈ 12.5–12.8 pp (not shown; see text)		

Wilson with n_{cells} underestimates uncertainty by ∼10% relative to bootstrap (widths 4.8 pp vs. 5.1 pp). Wilson with n_{eff} virtually matches the bootstrap (widths 5.3 pp vs. 5.1 pp); the two methods agree to within 0.1 pp for every model. Wilson with n_{studies} overestimates uncertainty by ∼2.5× because it discards the domain-level information and treats each study as a single binary observation.

Why bootstrap was adopted for all metrics

The numerical equivalence between Wilson n_{eff} and bootstrap for accuracy confirms that the low ICC ($\hat{\rho} = 0.044$) makes the independence assumption nearly tenable. Nevertheless, bootstrap was used uniformly for three reasons. First, Macro-F1 and Gwet's AC1 are composite statistics with no closed-form variance expression; bootstrap applies without modification. Second, the paired-bootstrap procedure used for inference-mode Δ values (Table 3) jointly resamples both conditions, correctly propagating correlation between them; Wilson intervals cannot be directly adapted for paired differences of non-proportion statistics. Third, a single method for all reported CIs ensures internal consistency and simplifies interpretation.